TIBER CAD

Trial Version Manual

Release 2.0

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CONTENTS

Part I

Getting Started

CHAPTER

ONE

INSTALLATION INSTRUCTIONS

In the following, VERSION denotes the version number of the TIBERCAD release you downloaded and INSTALLPATH denotes the directory where TIBERCAD gets installed. Version 2.3.0 of **GMSH** (http://www.geuz.org/gmsh) will be installed together with TIBERCAD. For the Linux version of GMSH you need OpenGL libraries installed on your system.

1.1 Prerequisites

Get the installer package for your OS/architecture from http://www.tibercad.org or by contacting support@tibercad.org. Table 1 lists the packages available for download. To run TIBERCAD you will also need a license file that you will have to copy into the installation directory of TIBERCAD.

In the Windows version, some graphical features such as graphical convergence monitors are only available if an X Window server is installed and running.

1.2 Windows installation procedure

To install TIBERCAD in Windows, run the setup program `tibercad-version_setup.exe` .

During the installation you can choose the installation directory. After finishing installation, copy your license file *(tibercad.lic)* into the

license subdirectory of the TIBERCAD

installation directory (INSTALLPATH/license), without changing its filename.

installer package name	*Target architecture*
tibercad-version_setup.exe	Windows 32-bit
tibercad-version_installer.bin	Linux 32-bit self-extracting installer

Table 1: Installer packages

1.3 Linux installation procedure

To install TIBERCAD in Linux, download and run the self-extracting installer `tibercad-version_installer.bin` and follow the installation instructions.

After installation, copy your license file *(tibercad.lic)* into the "license" subdirectory of the TIBERCAD installation directory (INSTALLPATH/license) without changing the filename. You can also provide the license file during installation.

The standard method to launch TIBERCAD is by means of a shell script that is installed alongside the TiberCAD executable. It takes care of setting all necessary environment variables.

If for some reason you have to run the executable directly, remember to set TIBERCADROOT to the TiberCAD installation directory (INSTALLPATH).

1.4 Quick start guide

In the "examples" subdirectory you can find several examples ready to run. They are the same as the tutorials on http://www.tibercad.org/documentation/tutorial/list.

1.4.1 Windows

Open Windows Explorer and go to the TIBERCAD installation directory. If you have write permission in the installation directory, you can browse to an examples directory and start the simulation by double clicking the input file, e.g. bulk.tib in *Step 1 - Modeling the device* . If not, copy the whole directory to a location in your personal area and run the examples from there.

If you cannot run TIBERCAD by double clicking an input file *(*.tib)* , then the input files are probably not correctly associated with the TIBERCAD executable. In this case, try to establish the association by right-clicking the input file, choosing `open with...`

> `>> Choose Program...` `>> Browse...` , browsing to the TIBERCAD installation directory

and choosing the TIBERCAD executable, tibercad.exe. A directory containing the simulation results will be created with the name provided in the input file.

1.4.2 Linux

After the correct installation of TIBERCAD you should be able to run TIBERCAD from the command line using the command tibercad. If not, you probably have to add the bin subdirectory of the TIBERCAD installation directory to your PATH environment variable or start the TIBERCAD executable using the absolute path (INSTALLPATH/bin/tibercad).

Copy the directory of the example you want to run, e.g. bulk Si, to your home directory or any place you have write permissions for. Change to the newly created directory and run TIBERCAD by (assuming *Step 1 - Modeling the device*)

```
$ tibercad bulk.tib
```

A directory containing the simulation results will be created with the name provided in the input file.

1.5 Bug reports / Feedback

Please send bug reports, feedback or suggestions to support@tibercad.org. When submitting bug reports, please always include the full version number of TIBERCAD you are running. The full version number appears in the first line of output when running the program:

$ tibercad

TiberCAD version 1.0.0-961

Usage: `tibercad` <inputfile>

CHAPTER

TWO

INPUT FOR TIBERCAD

Input for TIBERCAD is composed by an input file e.g. “input.tib” and a mesh file generated by a mesher software: as for now, mesh files from GMSH *(*.msh, v.1 and v.2.0)* and from ISE-TCAD *(*.grd)* are supported. Be sure that the material files are in the correct directory . To run the program, type: **tibercad** *input_file_name*

2.1 Description of Input file structure

A valid input file for TIBERCAD is a text file with the structure described in the following. In the whole input file, everything following a “#” is considered as a comment and is disregarded; blank lines can be present anywhere and are disregarded too. Input file is composed by several **blocks** :

A block is enclosed between **“{“** and **“}”** brackets and may include one or more blocks. Each block has a header made of one or two keywords. Each block may contain zero or any number of **parameter assignments** in the form:

” *tagname* = *tagvalue”* , where

- *“tagname”* is a string
- *“tagvalue”* is a single numerical or string item or a list of items between “(” and ”)”

parenthesis and separated by commas. e.g. (*cathode, anode*)

Format is free for the parameter assignments, provided that they are separated by spaces. Everything which follows a “#” is considered as a comment and is disregarded. For example:

```
electron_mobility field_dependent
  {
   region = buffer
   # type = doping_dependent
   low_field_model = constant
  }
```

Here and in the whole input file a string item can include a combination of characters, special characters and numbers, except spaces; if a space is found, the string item is taken as terminated.

The input file is composed by the following main classes of blocks:

Device, Module, Simulation

which will be described in the following.

2.2 Device section

```
Device hemt1
  {
   meshfile = hemt_msh.grd
   Region buffer
     {
      .........
     Doping
        {
         .........
        }
     }
   Region barrier_1
     {
      ............
     }
  ...................
  }
```

Device section includes the geometrical description of the device to be simulated; an optional *device name* can be associated to the Device object, e.g. *Device hemt1* .

In Device section, two kinds of block can be present: the Region block and the Cluster block. Outside of these blocks, general options common to all the device can

be defined. The most important is the mesh file definition:

meshfile : name of mesh file. N.B.: the extension is mandatory! (*.grd* for ISE-TCAD,

.msh for GMSH mesh file v.1 and v.2.0)

The definition of the mesh file associated to this simulation is compulsory:

```
meshfile = hemt_msh.grd
```

The **Region** blocks contain the description of the device in continuous media approach; the **Cluster** blocks define each a group of regions (mesh regions) even with different physical properties, but to be treated together somewhere in the simulation (e.g. quantum calculation). In this way it is possible to refer to the set of these regions simply by the **Cluster** name.

Each **Region** block must be preceded by the keyword **"Region"** , followed by the (single-word) name of the **TiberCAD Region** . The name of the **TiberCAD Region** can coincide with the name of a mesh region, as defined during the modeling of the device; in this case, if the keyword *mesh_regions* is absent, the **TiberCAD Region** will be associated to the mesh region identified by the name assigned to the **TiberCAD Region** . Otherwise, the **TiberCAD Region** will be associated to the mesh regions specified by the keyword *mesh_regions* .

```
Region QWell
  {
   mesh_regions = (well1,well2)
   structure = wz
   y-growth-direction = (1,0,-1,0)
   z-growth-direction = (-1,2,-1,0)
   x-growth-direction = (0,0,0,1)
   material = GaN
   Doping
     {
      density = 1e17
      type = donor
      level = 0.025
     }
  }
```

Here are the description of the available keywords for a **Region** block.

material (mandatory): name of the material associated to the present region, e.g Si;

it may be a ternary alloy, e.g AlGaAs, in this case keyword x described in the following has to be present.

x : alloy concentration, expressed as the molar fraction of the first component of the alloy; e.g. to express an alloy $Al_xGa_{1-x}As$ with molar fraction x = 0.2, that is $Al_{0.2}Ga_{0.8}As$, we select **AlGaAs** for the keyword material, and 0.2 for the keyword x, thus we write x = 0.2.

`mesh_regions` : (a list of) region physical name(s) as specified in the meshing program.

`structure` : crystal structure (wz = wurtzite, zb = zincblend)

`x-growth-direction, y-growth-direction, z-growth-direction` : Bravais vectors with Miller

indexes for wurtzite crystal (4 element vectors) or zincblende crystal (3 element vectors).

 A common crystal structure and growth directions definition may be applied to all the Regions of the Device, just by defining them everywhere in the Device block, but outside any specific Region block.

Subblock doping, with the keywords:

density : doping concentration [cm-3]

type : donor or acceptor

level : energy level of the dopant [eV]

```
Doping
  {
   density = 1e21
   type = donor
   level = 0.026
  }
```

Each **Cluster** block must be preceded by the keyword **"Cluster"** , followed by the (single-word) name of the **Cluster** .

```
Cluster Quantum_1
  {
   regions = (well, barrier1,barrier2)
  }
```

regions (mandatory): list of physical regions as specified in the meshing program, or TIBERCAD region names to be grouped in the cluster.

Regions and **Clusters** represent the macroscopical description of the device or structure

to be be simulated in **TiberCAD** . In the rest of the input file, the physical regions associated to the **Modules** will be indicated by means of the **TiberCAD Region** and **Cluster** names.

2.3 Modules

```
Module driftdiffusion
  {
   name = dd
```

```
   #regions = all
   statistics = FermiDirac
   strain_simulation = macrostrain
   plot = (Ec, Ev, Eg, eQFermi, hQFermi, eDensity, hDensity, Polarization,
   eCurrentDensity, hCurrentDensity,CurrentDensity, ContactCurrents,
   ElPotential, ElField, eMobility, hMobility, eConductivity, hConductivity)
    ..........
   Solver
     {
      nonlinear_solver = tiber
      nonlin_max_it = 15
      nonlin_rel_tol = 1e-12
      #nonlin_abs_tol = 1e-15
      nonlin_step_tol = 1e-5
     }
   Physics
     {
      polarization (piezo, pyro) {}
      mobility
        {
         type = field_dependent
         low_field_model = doping_dependent
        }
     }
Contact cathode
  {
   voltage = $Vd
  }
Contact anode
  {
   voltage = 0.0
  }
}
```

One or more module-blocks may be present: each module-block must be preceded by the keyword **"Module"** , followed by the (single-word) module name. This must be the name of one of the TIBERCAD modules. Here are the Modules implemented until now:

- `driftdiffusion` : Poisson-driftdiffusion transport of electrons and holes
- `thermal` : Heat balance simulation
- `excitontransport` : Exciton transport model
- `macrostrain` : Calculation of Elastic deformations in heterostructures
- `efaschroedinger` : Envelop Function Approximation (EFA) solution of single particle

Schroedinger equation for electrons and holes

- `quantumdensity` : Calculation of quantum density of electrons and holes.
- `quantumdispersion` : Dispersion of quantized states in k space
- `opticskp` : Optical properties (optical kp matrix elements)
- `opticalspectrum` : Emission spectrum (with k-space integration)
- `Sweep` : Parameterized execution of a module simulation (e.g. for the calculation

of output current characteristics)

- `Selfconsistent` : coupled calculations of different simulation modules.
- `DSC` : Simulation of a DSC solar cell1.

Each module-block usually contains a list of general options, such as plot and others specific to each module. Then, two main blocks define the Physics and the Solver models and parameters for this module.

Solver contains the solver parameters; depending on the Module, it can contain a

LinearSolver definition subblock.

Physics usually contains the definition of the physical models used in the simulation.

For example, for driftdiffusion module, recombination, electron mobility, trap, polarization and so on. A particular model is the Boundary model, which has an alias Contact for driftdiffusion module. The declaration of these models obey to the following syntax:

model_keyword type_specifier <block>

where *model_keyword* is the name of the physical model to be declared (e.g. trap model) and **type_specifier** is the name of a particular one among the available descriptions for that model. For example, for a trap model of **type** *acceptor*

```
trap acceptor
  {
   region = buffer
   Nt = 7e16
   Et = 0.5
   reference = cb
  }
```

An alternative multiple declaration is possible if no parameters, beyond default, are declared:

model_keyword (type specifier1, type specifier2,...)

```
recombination (srh, direct, auger) { }
```

In this example, several recombination models are defined (srh, auger, direct) each one with default parameters. For a detailed description of the models, please refer to the reference guide. Two special Modules are: the Sweep Module and the Selfconsistent Module

2.3.1 Module sweep

```
Module sweep
  {
   name = sweep_drain
   solve = driftdiffusion
   variable = $Vd
   start = 0.0
   stop = 2.0
   steps = 20
   plot_data = true
  }
Module sweep
  {
   name = sweep_gate
   solve = sweep_drain
   variable = $Vg
   start = -0.1
   stop = 1.0
   steps = 11
  }
```

Each sweep Module defines a set of calculations applied to a boundary region (e.g. a set of bias values to be assigned to a drain contact of a MOSFET for the calculation of an output drain IV characteristic), in this Guide referred to as sweep calculation.

The following keywords are defined for this feature:

`variable` : name of the variable to which the sweep is applied:

E.g.: variable = $Vg

indicates that the values are applied to the variable Vg, which is a quantity defined in a **Contact** definition

`start, stop, steps` : sweep starts from start value, is repeated steps times and stops

in stop

`solve` : name of the simulation (module) associated to the sweep calculation; it may

be the name of another sweep defined in the same block.

`plotvariable` (obsolete): specify the integrated quantity to be calculated during the

sweep and that will be shown in the output file `sweep_modelname_sweepvariable.dat` , eg. sweep driftdiffusion Vb.dat for a sweep of current calculation on the variable Vb (typically a contact voltage).

`plot_data` : default is false; if it is set to true, then output data will be written for each step of the sweep calculation, otherwise just the results for the final step will be present in the output.

Once a *sweep* calculation has been defined, it is treated as a special case of simulation and may be executed as an usual simulation: by adding it in the solve list, e.g. `solve = sweep_drain` .

2.3.2 Module selfconsistent

In this Module it is possible to define a self-consistent calculation based on two different simulation modules (e.g. driftdiffusion and excitontransport).

```
Module selfconsistent
  {
   name = sc_all
   solve = (tb, dens_el, dens_hl, dd)
   max_iterations = 5
   abs_tolerance = 1e-3
   rel_tolerance = 1e-6
  }
```

In **solve** the list of simulations to be performed self-consistently is specified. Now it is possible to execute the specified simulations in self consistent way, by using the **selfconsistent** keyword like a simulation name, in any **solve** assignment, e.g. solve = selfconsistent in **Simulation** section, or even in a **sweep** section.

2.4 Simulation section

In this block one can specify several general parameters and settings for the actual calculation to be run, such as the temperature, the process-flow of simulation, etc.

`searchpath` : path for material files

`mesh units` : units of measurements used in the meshing (relative to meters): e.g., 10^{-6} for μm

`dimension` : dimension of simulation (1,2,3)

`temperature` : temperature of the system [K]

`solve` : list of simulations to be executed, in the order of execution; if the list contains "sweep", a sweep is performed as specified in sweep block in the Solver section.

```
solve = (strain,driftdiffusion,
quantum_electrons, quantum_holes)
```

`resultpath` : path for output directory

`output format` : format of the output data: *gmv* for **GMV** , ise for **Tecplot** , grace for

xmgr (ascii data column type), *vtk* for **Paraview** .

`plot` : list of output variables which are calculated and available in output files. It

may be overriden by defining list specific to one or more modules.

2.5 Output description

At the end of the execution, the program will write the results of the simulation in the directory specified by resultpath , with the format specified by output format. The output variables are specified in a list plot, in each Module.

TiberCAD output is divided in two classes: **mesh-based** and **mesh-independent** quantities.

Mesh-based quantities are all the quantities associated with the nodes of the mesh, such as Fermi level, electron and hole density, conduction and valence band, etc., together with all the quantities associated with the elements of the mesh, such as current density.

The output values for these quantities are reported in the files simname msh.ext, where *simname* is the simulation module used for the calculations and ext is the extension of the chosen file format, e.g. vtu for paraview output.

```
strain_msh.vtu
```

In the case a sweep calculation is performed and the plot data keyword is set to true, the output files are of the kind simname sweepvariable step value msh.ext, where

sweepvariable is the variable with respect to which the sweep is performed (e.g. gate

voltage) and step value is the value of this variable at that step; e.g the result at the step *Vbias = 1.1* will be found in the file:

```
dd_Vbias_1.1_msh.vtu
```

`mesh-independent` quantities are the quantities which are not associated to the mesh, for example current at the contacts of a diode or quantized energy levels in a quantum well. These **mesh-independent** quantities are displayed in separated files, with the format simname.ext, e.g `quantum_electrons.dat` , where simname is the name of the model (simulation) associated to the results. If a sweep is performed, the output file gets the format sweep name simname.ext, where *sweep_name* is the name of the sweep performed, for example

```
sweep_drain_driftdiffusion.dat
```

Inside the file, output values for all the steps of calculation are shown.

2.6 Example of Input file

Here is an example of the input file template:

```
Device hemt1
  {
   meshfile = hemt_msh.grd
   mesh_units = 1e-6
   material = GaN
   y-growth-direction = (0,0,0,1)
   z-growth-direction = (1,0,-1,0)
   x-growth-direction = (-1,2,-1,0)
   Region barrier
     {
      material = AlGaN
      x = 0.14
     }
   Region barrier_doped
     {
      mesh_regions = (barrier_dop_s, barrier_dop_d)
      material = AlGaN
      x = 0.14
      Doping
                {
         density = 1e21
         type = donor
         level = 0.026
        }
     }
  Region buffer { }

      Region buffer_doped
    {
     mesh_regions = (buffer_dop_s, buffer_dop_d)
     Doping
           {
        density = 1e21
        type = donor
        level = 0.026
       }
    }
  Region cap
    {
     Doping
           {
        density = 5e18
        type = donor
```

```
        level = 0.026
       }
    }
  Region cap_doped
    {
     mesh_regions = (cap_dop_s, cap_dop_d)
     Doping
           {
        density = 1e21
        type = donor
        level = 0.026
       }
    }
  Region passivation
    {
     mesh_regions = (passiv1, passiv2, passiv3, passiv4)
     material = SiN
    }
  Cluster semiconductor
    {
     regions = -passivation
    }
  }
Module driftdiffusion
  {
   name = dd
   regions = all
   coupling = electrons
   save_state = true
   #load_state = output_Nt7e16_out/dd_Vd_50.tsv
   statistics = FermiDirac
   strain_simulation = macrostrain
   plot = (Ec, Ev, Eg, eQFermi, hQFermi, eDensity, hDensity, Polarization,
           eCurrentDensity, hCurrentDensity,CurrentDensity, ContactCurrents,
           ElPotential, ElField, eMobility, hMobility, eConductivity, hConductivity

      NonlinearSolver linesearch
     {
      # type = ls
      relative_tolerance = 1e-15
      step_tolerance = 5e-3
      max_iterations = 15
      LinearSolver petsc
        {
         ksp_type = pconly
         preconditioner = lu
        }
```

```
    }
   Physics
    {
     recombination (srh, direct, auger) { }
     electron_mobility field_dependent
       {
        region = buffer
        low_field_model = constant
       }
     electron_mobility doping_dependent
       {
        region = (barrier, barrier_doped, cap, cap_doped, buffer_doped)
       }
     trap fixed_charge
       {
        region = GaN/SiN
        Nt = 2.74e13
       }
     trap acceptor
       {
        region = buffer
        Nt = 7e16
        Et = 0.5
        reference = cb
       }
     polarization (piezo, pyro) { }
    }

  Contact drain
    {
     voltage = @Vd
    }
  Contact source
    {
     voltage = 0.0
    }
  Contact gate
    {
     regions = (gate1, gate2)
     type = schottky
     work_function = 1.5272
     voltage = @Vg
    }
  }

Module macrostrain
  {
```

```
   regions = -passivation # all the device except Region "passivation"
   plot = all
   LinearSolver # default
     {
      tolerance = 1e-7
      max_iterations = 10000
      #xmonitor = true
      pc = composite
     }
   Physics
     {
      Boundary substrate
        {
         regions = bottom
         material = GaN
         y-growth-direction = (0,0,0,1)
         z-growth-direction = (1,0,-1,0)
         x-growth-direction = (-1,2,-1,0)
        }

      Boundary extended_material
        {
         regions = side
        }
     }
  }

Module sweep
  {
   name = sweep_g
   variable = Vg
   solve = sweep_d
   start = -1
   stop = 1.0
   steps = 2
   max_step = 0.25
   min_step = 1e-4
   #plot_data = true
  }

Module sweep
  {
   name = sweep_d
   variable = Vd
   solve = driftdiffusion
   start = 0.0
   stop = 50.0
```

```
   steps = 200
   min_step = 1e-4
   plot_data = true
  }

Module selfconsistent
  {
   name = relaxation
   solve = (macrostrain, driftdiffusion)
   absolute_tolerance = 1e-3
   relative_tolerance = 1e-5
  }

Simulation
  {
   temperature = 300
   verbose = 3
   solve = (macrostrain, dd, sweep_g)
   logfile = output_Nt5e16_Al0.215_SiN/hemt.log
   # these parameters can be defined in modules, too
   resultpath = output_Nt5e16_Al0.215_SiN
   output_format = vtk
  }
```

CHAPTER

THREE

ELASTICITY

3.1 Abstract

Elasticity is a Finite Elements solver for mechanical equilibrium problems. It brings features typically developed for structural stability solvers into device modeling. The coupled treatment of the electro-mechanical problem within a unique framework results very useful to explore for multidisciplinary ideas.

3.2 Mini theoretical intro

Assuming a small displacement, Elasticity elasticity computed the strain by solving the equation

$$\frac{\partial}{\partial x_j} C_{ijlk} \frac{\partial u_l}{\partial x_k} = f_i$$

where C is the stiffness constant and f is the total external body force. The latter may include several contribution such as the converse piezoelectric effect the thermal expansion and a lattice mismatch induced strain. The latter, described in the example below, occurs when an interface is created between two materials with different lattice constants. Let ϵ^{LM} be the lattice mismatch strain, then the effective body force reads as :

$$f_i = -\frac{\partial}{\partial x_j} C_{ijlk} \epsilon^{LM}_{lk}$$

3.3 Example

In the following we will compute the strain induced by the lattice mismatch in a system comprising a layer of GaN between two contacts of $Al_xGa_{1-x}N$. Once the mesh is created with gmsh (dis-

tributed along TiberCAD) we may start to build the input file. First, we have to insert the region section

```
Device
  {
   dimension = 3
   meshfile = mesh.msh
   mesh_units = 1e-9
   material = AlGaN
   x = 0.2
   x-growth-direction = (-1,0,1,0)
   y-growth-direction = (-1,2,-1,0)
   z-growth-direction = (0,0,0,1)
   Region Well  {material = GaN}
  }
```

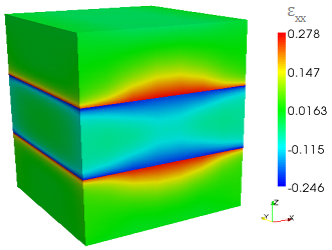
All the options included in this section, such ad the material, axis growth and alloy concentration, apply for the whole structure. If we want to specify a region with different properties, we may use the keyword Region and refer to a region name, as specified in the mesh file.

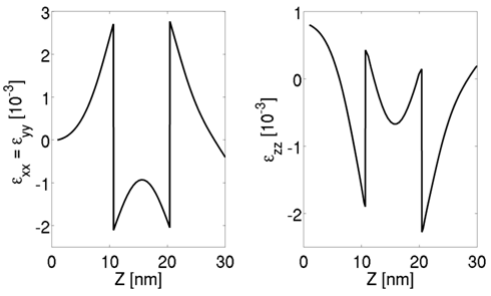
The second part of the input file is devoted to the module declaration

```
Module elasticity
  {
   Physics
     {
      BodyForcelattice_mismatch
        {
         x-growth-direction = (-1,0,1,0)
         y-growth-direction = (-1,2,-1,0)
         z-growth-direction = (0,0,0,1)
         reference_material = AlGaN
         x = 0.2
        }
      Contact Base {type = clamp}
     }
  }
```

In physics we declare the physical models to be applied to our device. In our case, with BodyForce lattice mismatch we are adding a body force into the system, induced by the lattice mismatch with the reference lattice which in this case is $Al_{0.2}Ga_{0.8}N$. In order to avoid a free standing device we may want to freeze a surface. With Boundary Clamp we fix, in this case, the region Base.

The third part is simply Simulation **{solve = elasticity}** where the default name *elasticity* is used. By default, files for 3D system are written in vtu which can be read from Paraview. Output data about strain are shown below.


ε_{xx}
0.278
0.147
0.0163
-0.115
-0.246


$\varepsilon_{xx} = \varepsilon_{yy}$ [10^{-3}]
Z [nm]
ε_{zz} [10^{-3}]
Z [nm]

CHAPTER

FOUR

DRIFT DIFFUSION

In this example we will see a very simple TiberCAD simulation:

1D calculation of Poisson and drift-diffusion for a bulk Silicon sample.

The following files should be in your working directory:

bulk.tib : **TiberCAD** input file

bulk.msh : mesh file

The mesh file can be obtained from the following GMSH geo file : bulk.geo

This is the summary of the Sections of this Tutorial :

4.1 Step 1 - Modeling the device

As a first step, we have to model the device. To do so, you can use DEVISE module of ISE-TCAD 9.5 software package or GMSH program.

Here we'll see in details the procedure for GMSH.

There are two possible ways to use GMSH:

Interactive, using the graphical interface Using a script file In the following we'll see how to write a basic GMSH script (bulk.geo); for any details please refer to GMSH manual GMSH (http://geuz.org/gmsh/).

 : In a **1D** simulation it is assumed that the geometrical model is restricted to the **x axis** . In a **2D** simulation it is assumed that the geometrical model is restricted to the **xy-plane (z=0)** Any other geometrical orientation could give unpredictable results

In a GMSH script, several variables can be defined and given a value in this way:

```
L = 1
d = 0.01
```

these are valid GMSH variables: L is just the length of the Si sample; d is the value of a *characteristic mesh length* (see below).

Definition of geometrical entities **Points**

```
Point(1) = {0, 0, 0, d}
Point(2) = {L, 0, 0, d}
```

The first three expressions inside the braces on the right hand side give the three X, Y and Z **coordinates** of the point; the last expression **(d)** sets the **characteristic mesh length** at that point, that is the *"size"* of a mesh element, defined as the length of the segment for a line segment, the radius of the circumscribed circle for a triangle and the radius of the circumscribed sphere for a tetrahedron.

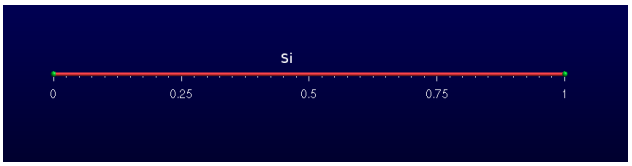
Thus, the smaller is the value of d, the greater is the mesh density close to that point.

The size of the mesh elements will then be computed in GMSH by linearly interpolating these characteristic lengths in the whole mesh.

Definition of a geometrical entity **Line**

```
Line(1) = {1, 2}
```

The two expressions inside the braces on the right hand side give the identification numbers of the start and end points of the line.



Definition of the physical entity **Physical Line bulk**

```
Physical Line("bulk") = {1}
```

The expression(s) inside the braces on the right hand side give the identification numbers of all the **geometrical lines** that need to be grouped inside the *physical line* .

In this way, in general, physical regions are created which associate together geometrical regions, and then the related mesh elements, which share some common physical properties. It's only these physical regions which can be referred to outside GMSH. In TiberCAD, this is done by associating one or more physical regions to a **TiberCAD region** through the keyword *mesh_regions* (see in the following).

Definition of two physical entities Physical Point:

```
Physical Point("anode2)   = {1}
Physical Point("cathode") = {2}
```


: In general, in a nD simulation, **(n-1)D** physical regions (points in 1D, lines in 2D, surfaces in 3D) are used by TiberCAD to impose the required boundary conditions.

Each (n-1)D physical region defined in this way in GMSH will be associated in TiberCAD to a boundary condition region, through the keyword **BC_reg_numb** . Thus, in this case, Physical points Anode and Cathode will be associated respectively to two **Contact** regions (see in the following).

4.2 Step 2 - Meshing the device

The *.geo* script file with the geometrical description can be run in GMSH, to display the modelled device and to mesh it through the GMSH graphical interface. Alternatively, a *non-interactive* mode is also available in GMSH, without graphical user interface.

For example, to mesh this 1D tutorial in non-interactive mode, just type:


gmsh bulk.geo -1 -o bulk.msh

where bulk.geo is the geometrical description of the device with GMSH syntax:

-1 means *1D mesh generation*

some command line options are:

-1 , -2, -3 to perform *1D, 2D or 3D* mesh generation,

-o **mesh_file.msh** to specify the name of the mesh file to be generated

In this way, a .msh has been generated and is ready to be read in TiberCAD.

4.3 Step 3 - TiberCAD Input file

Now we have to write down the **TiberCAD input file** (bulk.tib). For a detailed reference see the user guide (Input file) and Getting started 1

4.3.1 Description of Device Regions

First, we have to list all the **TiberCAD Regions** present in our Device: a TiberCAD Region is usually a section of the device featuring the same material and possibly the same doping.

```
Device
  {
   meshfile = bulk.msh

   Region bulk
     {
      material = Si

      Doping
        {
         Nd = 1e16
         type = donor
        }
     }
  }
```

The TiberCAD Region bulk is made of Silicon and n-doped with a concentration 1 x $10^{16} cm^{-3}$.

Through the keyword **Region** , one GMSH physical region (Physical Lines in 1D, Physical Surfaces in 2D, Physical Volumes in 3D) previously defined in the GMSH mesh (*Step 1 - Modeling the device*), can be associated to the present TiberCAD Region, in this way:

```
Region  GMSH_physical_region_name
```

In this case, the Physical Line bulk is associated to the TiberCAD Region bulk.

Alternatively, through the optional keyword **mesh_regions** , one or more GMSH physical regions can be associated to a single TiberCAD Region .

4.3.2 Definition of Simulation

Now we define the **Simulation** driftdiffusion_1: it belongs to the class **driftdiffusion**

```
Module driftdiffusion
  {
   name = driftdiffusion  # this is the default name

   regions = all          # 'all' is the default

   # what we want to plot
   plot = (Ec, Ev, eQFermi, hQFermi, ContactCurrent)
   ....
```

The TiberCAD simulation *driftdiffusion_1* , belonging to the model driftdiffusion, will be applied to the whole device structure (`regions = all`)

4.3.3 Definition of Boundary Conditions

The anode and cathode contacts of our 1D Si sample are defined as **Boundary conditions regions** (`Contact anode, Contact cathode`) in the following way:

Module driftdiffusion {

name = driftdiffusion

regions = all

plot = (Ec, Ev, eQFermi, hQFermi, ContactCurrent)

Contact anode { voltage = $Vb } Contact cathode { }

}

Through the keyword Contact , one (n-1) -dimension GMSH physical region (Physical Point in 1D, Physical Line in 2D, Physical Surface in 3D) previously defined in the GMSH mesh (*Step 1 - Modeling the device*), can be associated to the present TiberCAD Contact, in this way:

Contact `GMSH_physical_region_name`

In this case, the *Physical Point* **anode** is associated to the TiberCAD Contact anode and the Physical Point cathode is associated to the TiberCAD Contact cathode.

Both contacts are defined as *ohmic* , cathode is assigned a fixed `voltage = 0.0` , while anode voltage is given by the value of the variable *Vb*

```
voltage = @Vb
```

4.3.4 Definition of Simulation parameters

Vb is specified in the sweep block, in the Solver section:

```
Module sweep
  {
   solve = driftdiffusion
   variable = $Vb
   start = 0.0
   stop = 1
   steps = 10
   plot_data = true
  }
```

In this way, the simulation `driftdiffusion_1` is performed for 10 (steps = 10) values of the anode voltage (variable = Vb), between 0 and 1.

For each step we want to plot the solution variables specified in the driftdiffusion module (`plot_data = true`).

4.3.5 Definition of Execution parameters

In the **Simulation** section , we decide *which* simulations to perform and in which *order*; set `solve = sweep` , to execute the sweep which run driftdiffusion_1 simulation for the specified loop.

```
Simulation
  {
   verbose = 2

   solve = sweep

   resultpath = output
   output_format = grace
  }
```

Output files with conduction and valence band profiles (plot = Ec,Ev..) and all the calculated values of the current at the contacts (*ContactCurrents*) (the IV characteristic) are generated.

To increases the amount of information written to the screen we can vary the verbose level (verbose = 2).

4.4 Step 4 - Run TiberCAD

Now we can run TiberCAD

by double clicking on bulk.tib file (in Windows)

or by command line in linux: tibercad bulk.tib

4.5 Output

The generated Output files are:

- `driftdiffusion_materials.dat` : material (mesh) regions, in this case just region 1
- `driftdiffusion_nodal.dat` : nodal quantities (here conduction and valence band)
- `sweep_driftdiffusion_Vb.dat` : integrated current at the two contacts for each sweep step

Attachment Size

- bulk.tib 1.16 KB
- bulk.geo 181 bytes
- bulk.msh 4.49 KB

CHAPTER

FIVE

THERMAL

5.1 Introduction

The steady state heat flux, within the diffusive approach, is given by the Fourier's law, i. e. $J_i = -\kappa_{ij}\frac{\partial T}{\partial x_j}$ where κ is thermal conductivity second-rank tensor (which is assumed temperature independent).

The power balance is ensured by the continuity equation for the thermal flux $\frac{\partial J_i}{\partial x_i} = H$ where H is the total heat source.

Thermal module computes the balance between the heat generated and the heat dissipated. Here we get the temperature profile of a rectangular device with an hotspot in the center.

The device block specifies the material (Silicon) and the physical regions. In this case we have the HotSpot and the Base region

```
Device
  {
   material = Si
   Region HotSpot{}
   Region Base{}
  }
```

Let's now declare the thermal module. The temperature is fixed at the left and right side of the domain. The hotspot is included only in the HotSpot region

```
Module thermal
  {
   Physics
     {
      Contact left
        {
         type = heat_reservoir
         temperature = 300
        }
```

```
      Contact right
        {
         type = heat_reservoir
         temperature = 300
        }
      HeatSource constant
        {
         regions = HotSpot
         H = 1e10
        }
    }
  }
```

In the simulation region we write which modules should be solved.

```
Simulation{solve = thermal}
```

The temperature map is shown below

5.2 Boundary conditions

The boundaries of the thermal simulation domain are set to ideally thermal insulator by default.

Dirichlet boundary conditions can be imposed with the keyword heat_reservoir.

Example:

```
Contactheat_reservoir
  {
   region = substrate
```

```
   temperature = 300
  }
```

The surface resistance can be added with the keyword boundary surface_resistance. (r_surf is in cm2 W/ K).

Example:

```
Contactsubstrate
  {
   type = surface_resistance
   r_surf = 0. 05
   temperature = 300
  }
```

5.3 Heat sources

The constant value heat source can be included with heat_source constant. The heat source must be in W/cm3.

Example:

```
HeatSourceconstant
  {
   region = qdot
   H = 1e10
  }
```

Heat generated by the Joule's effect is included with heat_source joule. The name of the drift-diffusion simulation is given by dd_simulation.

Example:

```
HeatSourcejoule
  {
   dd_simulation = dd
  }
```

CHAPTER

SIX

QUANTUM EFA CALCULATIONS

In TiberCAD, it is possible to perform quantum calculations in the framework of Envelope Function Approximation (EFA): eigenstates and eigenfunctions of a given system, dispersion of quantum states and particle quantum density can be obtained respectively by means of the following Modules:

- Module efaschroedinger
- Module quantumdispersion
- Module quantumdensity

The optical properties are calculated by the following modules

- Module opticskp
- Module opticalspectrum

6.1 Module efaschroedinger

The EFA calculation of eigenstates and eigenfunctions are performed by the **Module** efaschroedinger. A typical example is the following:

```
Module efaschroedinger
  {
   particle = el
   poisson_model_name = dd
   strain_model_name = strain # macrostrain
   name = quantum_el
   regions = quantum
   plot = (EigenFunctions, EigenEnergy, EnergyLevels)
   Solver
     {
      number_of_eigenstates = 10 # 30
     }
```

```
  Physics
    {
     model = conduction_band
    }
}
```

6.1.1 Module options

The following options infiuence the behaviour of the Module efaschroedinger:

`particle = string` defines for which particle (electron or hole) Schroedinger equation is solved. Possible values are el and hl. A different Module efaschroedinger has to be defined for each particle to be solved.

`poisson_model_name = string` defines the name of the simulation (Module driftdiffu- sion) that can provide electric potential

`strain_model_name = string` defines the name of the simulation (Module macrostrain) that can provide elastic strain

`regions = string` defines the regions associated to this EFA simulation

6.1.2 Solver section

The Solver section of the Module efaschroedinger contains the following options:

`number_of_eigenstates = integer` defines the number of eigenvalues and eigenfunctions to be found.

`Dirichlet_bc_everywhere = boolean` : if true (default value), Dirichlet boundary

conditions are imposed over all the boundaries of the simulation region

`solver = string` : defines the solver for the eigenvalue problem, possible values are:

arnoldi, lapack, krylovshur. The default value is **krylovshur** . In the case of the lapack solver all the eigenvalues are computed. In the case of arnoldi or krylovshur solver it is necessary to specify which and how many eigenvalues have to be computed. The idea is that the iterative solver calculates several eigenvalues that are close to a specific number, refered to as the *spectral_shift*

`max_iteration_number = integer` : maximum number of iteration, used as a stop condition

`eigen_solver_tolerance = double` :numerical eigensolver tolerance used as a convergence criteria

`spectral_shift = double` :the closest eigenvalues to this value (eV) are found. If not

defined, then it will be calculated internally from the band edges.

`ksp_type = string` : Krylov subspace method type; bcgsl, gmres, cg

`pc_type = string` : preconditioner type: cholesky, jacobi, ilu , composite.

6.1.3 Physics section

`model = string` : possible values are : conduction band , for single conduction band

model (Γ point) ; kp for $\mathbf{k} \cdot \mathbf{p}$ model

`kp_model = string` : possible values are : 6?6 , 8?8.

6.1.4 Output

The available output variables, to be specified in the plot option, are the following:

- EigenEnergy Eigen energy in eV
- EigenFunctions $|\psi(\mathbf{r})|^2$ function of the eigenstate
- Occupation probability to find the state occupied. It is calculated assuming Fermi

distribution and mean electrochemical potential and temperature:

- EnergyLevels graphical output used for showing the energy level over the band

diagram.

6.2 Module quantumdispersion

With the Module quantumdispersion it is possible to calculate the dependence of quan- tum eigenstates on **k** -vector. Such dependence gives the *quantum state dispersion* . The simulation name is **quantumdispersion** .

```
Module quantumdispersion
  {
   simulation_name = dispersion1D_el
   regions = all
   quantum_simulation = quantum_el
   min_eigenvalue_number = 0
   max_eigenvalue_number = 5
```

```
  wedge = half
  k_space_dimension = 1
  k1 = (0, 0.1, 0)
  number_of_nodes = (10)
  output_format = grace
  plot = k-space_dispersion
 }
```

The dispersion of quantum states is calculated at k-points that are nodes of the mesh in k-space.

The main parameters are:

- **quantum simulation** : name of the efaschroedinger simulation.
- **min eigenvalue number** , **max eigenvalue number** : the dispersion is calculated

for the states number *i* , where

max_eigenvalue_number $\geq i \geq$ min_eigenvalue_number

The rest of the parameters (wedge, k space dimension, etc...) define the k-space.

6.2.1 Output

The output variable name is **k-space_dispersion** . The output format for the dispersion can be controlled independently of the general specification in the `Simulation` section by redefining the `output_format` keyword.

6.3 Module quantumdensity

In Module quantumdensity you can define the calculation of particle (electron,hole) den- sity, based on the result of a previous calculation of the system eigenstates (e.g. with efaschroedinger module).

```
Module quantumdensity
  {
   name = dens_el
   regions = quantum
   plot = quantum_density
   k-space_dimension = 0
   k-space_basis = true
   k1 = (0, 0, 0.1)
   #number_of_nodes = (4)
   #wedge = half
   quantum_simulation = quantum_el
   degeneracy = 2
```

```
   refine_fraction = 0.20
   relative_accuracy = 0.01
   refine_k_space = true
   output_density_in_k_space = true
   uniform_refinement = false
   mesh_order = FIRST
   first_state = 3
   initial_eigenstates_number = 10
   analytic = true
  }
```

The available options are:

- **quantum_simulation** : name of the efaschroedinger simulation.
- degeneracy: degeneracy of the quantum state
- **initial_eigenstates_number** : initial number of eigenstates for the Schroedinger

equation

- **analytic** = { true | false } If true then the density is calculated analytically,

otherwise numerically.

6.3.1 Output

The output parameter is **quantum_density**.

6.4 Module opticskp

By defining the Module **opticskp** , calculation of optical properties is enabled.

```
Module opticskp
  {
   name = optics
   regions = quantum
   plot = (optical_spectrum_k_0 )
   initial_state_model = quantum_el
   final_state_model = quantum_hl
   initial_eigenstates = (0, 9)
   final_eigenstates = (0, 15)
   polarization = (0, 0, 1)
   Emin = 2.8
   Emax = 3.6
   dE = 0.001
  }
```

Here, *initial_state_model* and *final_state_model* are, respectively, the quantum simulations (**efaschroedinger** module) associated respectively to the initial state of optical transition (e.g. electron), and to the final state of optical transition (e.g. hole). *initial_eigenstates* and *final_eigenstates* refer to the range of eigenstates to be taken in account for optical calculations.

By specifying a range of energy values in this way:

```
Emin = 3.0
Emax = 5.0
dE = 0.001
```

the emission optical spectrum for **k=0** is calculated.

6.4.1 Output

The output variables for optics calculations are:

- **optical_spectrum_k_0** : optical emission spectrum for *k=0*.

6.5 Module opticalspectrum

By defining the Module **opticalspectrum** , optical matrix elements are used to calculate the associated (emission) spectrum with a k-space integration.

```
Module opticalspectrum
  {
   k_space_dimension = 2
   k-space_basis = true
   k1 = (0, 0, 0.1)
   k2 = (0, 0.1, 0)
   refine_fraction = 0.30
   relative_accuracy = 0.01
   refine_k_space = true
   number_of_nodes = (2, 2)
   wedge = quarter
   plot = (optical_spectrum)
   optical_matr_elem_model = opticskp
   polarization = (0, 0, 1)
   Emin = 3.0
   Emax = 5.0
   dE = 0.001
  }
```

The parameters are the following:

`k_space_dimension` = **1** for 2D simulations, **2** for 1D simulations. k-space basis is

true if the k-space is defined by means of k-vectors; if **false** , vectors are expressed in real space

If `refine_k_space` = **true** , that is adaptive k-mesh refinement is enabled, all the elements whose error is greater than the value (1-refine fraction)* (maximum error) are going to be refined. In this case, Error is just the integrated quantity. The refinement will end when the *relative_accuracy* is obtained.

`number_of_nodes` = numb. of elements in k mesh, along each direction

`wedge` = half | quarter, to reduce calculation time, by exploiting symmetry.

`optical_matr_elem_model` = name of the *opticskp* model associated

`polarization` = light polarization (vector)

`Emin, Emax, dE` : energy range and step of spectrum calculation.

6.5.1 Output

The output variables for optics calculations are:

- **optical_spectrum** : k-space integrated optical emission spectrum.

CHAPTER

SEVEN

SIMULATION DYE SOLAR CELLS

The DSC model is tagged as *dssc* . In **options** subsection we indicate the simulation name:

```
model dssc
  {
   options
     {
      simulation_name = <name_of_the_model>
      plot(Potential, Density, Current, ContactCurrents)
     }
   BC_regions
     {
      ...
     }
  }
```

The boundary conditions are inserted in the **Model** section, the two contacts are the photoanode and the cathode. The photoanode must be the boundary of a region where $\mathbf{TiO_2}$ is present, on the contrary the cathode must be the boundary of a region where the **electrolyte** is present.

```
BC_regions
  {
   BC_region <name_of_the_anode>
     {
      type = ohmic
      bias = @V[0.0]
     }
   BC_region <name_of_the_cathode>
     {
      type = Pt
      load = @R[1e8]
     }
  }
```

The anode is an ohmic contact, it contains only the sweep of the bias applied for the electrochemical potential of the electrons. The cathode must contain the initial external resistance (**load = R**).

If a simulation of the cell under illumination is made the initial value of *load* must be set to a high value, in the range of $10^8 \omega cm^2$ in order to have no current during the light intensity sweep. In case instead the simulation performed is under dark conditions where an external bias only is applied `load = 1.0`.

The generation term is related to the flux of photons which reaches the active TiO_2 regions and the dye present in the cell. We assume a simple Beer-Lambert exponential decay for charge generation of the form:

$$G(\mathbf{r}) = \int_{\lambda_1}^{\lambda_2} \alpha(\lambda)\Phi(\lambda)e^{-\alpha(\lambda)\hat{n}\cdot\mathbf{r}}d\lambda \tag{7.1}$$

where α is the absorption coefficient (in μ^{-1}) of the chosen Dye, $\Phi(\lambda)$ the intensity of the light power at that wavelength for the spectrum of the sun. The part of the input file for the generation model:

```
model dssc_generation
  {
   options
     {
      regions = (<TiO2_region_name_1>, <TiO2_region_name_2>, ...)
      plot = (Distance, Generation)
      light_direction = <vector_indicating_the_direction_of_light>
     (example, illumination from x direction: light_direction = (1, 0, 0))
      light_intensity = @x
      dye = N719
     }
   BC_Regions
     {
      BC_Region <name_of_the_boundary>
        {
        }
     }
  }
```

In the generation input flle must be specified:

- (**regions**) the regions where we want that there is generation (where the Dye is present);
- (**plot**) if we want to plot the generation;
- (**light_direction**) the vector which fixes the direction from where the light comes;
- (**light_intensity**) the light intensity;
- (**light_intensity**) the light intensity;
- (**dye**) the dye used in the cell;

The light intensity is defined in a sweep (see section *Solver*). It can be set to reach 1 that means one Sun, or a larger or smaller illumination intensity (0.1, 2.0, etc.). There is another flag called

illumination_spectrum that can be used if we want to change the spectrum profile F with another illumination source (for example if we assume the cell under concentration of light). The file used by default for Φ is the standard 1.5 AM spectrum of the sun contained in the material database in the file `Sun1p5am` .

The last part of the generation is the definition of the boundary. This boundary says to the code which is the boundary region of the grid from where the light penetrates in the device. The information given by the boundary and the light direction vector defines the coming of light and direction of rays.

7.1 Device

Device section for a DSC simulation:

```
Device
  {
   Region <name of the porous region>
     {
      mat = TiO2mes
      porosity = 0.5
     }
   Region <name of the electrolyte region>
     {
      mat = TiO2mes
      TiO2 = false
     }
  }
```

For every region of the device it is specified the material file from the database (the TiO2mes contains standard parameters for both TiO_2 and electrolyte, so it can be used for both regions). For every region must be specified if it contains TiO_2, or electrolyte or both.

This can be done setting two flags called `TiO2` and `electrolyte` . If set true the material is present in the region, if set **false** it is not present. By default they are assumed both **true** (porous region). In the second region of the example showed there is electrolyte only and `TiO2` = **false** is explicitly specified. In case both materials are present (porous region) a porosity must be specified (in the range between 0, TiO_2 only, and 1, electrolyte only).

If one material is not present the porosity is automatically set to 0 or 1.

7.2 Physics

The set of parameters that can be defined for the entire device are enlisted in table 3.1. For the generation:

```
dssc
  {
   generation = dssc_generation
  }
```

7.3 Solver

Three sweeps are needed for the plot of the entire I-V under illumination. The first sweep is needed to make the transition from dark conditions to full open-circuit conditions under

Parameters

	Poisson equation	
porosity	Porosity of porous material	
perm_oxide	Relative perm. of the TiO_2	
perm_electrolyte	Relative perm. of the electrolyte	
k_3	Oxidized Dye regeneration rate constant	s^{-1}
	Drift Diffusion equation	
ne	Dark electron density	cm^{-3}
nI	Dark iodide density	mol/l
nI3	Dark electron density	mol/l
mu_e	Electron mobility	cm^2/Vs
D_I	Iodide diffusion constant	cm^2/s
D_I3	Triiodide diffusion constant	cm^2/s
D_C	Cation diffusion constant	cm^2/s
	Recombination term	
k_e	Recombination constant rate	s^{-1}
beta	Non-linearity exponent in the electron density recombination	

Table 7.1: Physical parameters that can be set for the model divided in subsets relative to different processes in the cell.

illumination. The high value of the load R maintains the current to zero. Then a second sweep is used to pass from open circuit condition to short circuit condition lowering the external load from a high external load to a small one (R = 1). Finally, the voltage sweep compute the I-V characteristics under illumination. In case of dark simulation (application of an external bias without illumination) the first and second sweep are not needed. The external load for the cathode can be set directly to R = 1.

```
Solver
  {
   Sweep
     {
      sweep_gen
```

```
        {
         simulation = (dssc_generation, dssc)
         variable = x
         values = (0, 1e-9, 1e-8, 1e-7, 1e-6, 1e-5, 1e-4, 1e-3, 1e-2, 0.1, 1)
         plot_data = true
        }
      sweep_R
        {
         simulation = dssc
         variable = R
         values = ( 1000, 100, 1)
         plot_data = true
         }
      sweep_V
        {
         simulation = dssc
         variable = V
         values = (0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0,8, 0.9, 1.0)
         plot_data = true
        }
      }
  }
```

The intensity of illumination can be changed sweeping the value of x different to 1 (1 = 1 Sun of power).

7.4 Output

The output that want to be plotted are enlisted in section Model within option. See *Table nodal* , *Table elemental* and *Table scalar* .

Nodal quantities

	`Potential`	
`eQFermi`	Electron electrochemical potential	eV ($-e\phi_n$)
`IQFermi`	Iodide electrochemical potential	eV ($-e\phi_{I^-}$)
`I3QFermi`	Triiodide electrochemical potential	eV ($-e\phi_{I_3^-}$)
`CQFermi`	Cation electrochemical potential	eV ($-e\phi_C$)
`ElPotential`	Electrostatic potential	eV
`Eredox`	Electrolyte electrochemical potential	eV ($-e\phi_{Red}$)
	`Density`	
`eDensity`	Electron density	cm^{-3}
`IDensity`	Iodide density	cm^{-3}
`I3Density`	Triiodide density	cm^{-3}
`CDensity`	Cation density	cm^{-3}
`Generation`	The net electron generation rate	$cm^{-3}s^{-1}$
`NetRecombination`	The net recombination rate	$cm^{-3}s^{-1}$

Table 7.2: Nodal quantities. The flags `Potential` and `Density` allow to plot all the electrochemical potentials and all the densities, including generation and recombination.

Elemental quantities

	`Current`	
`ElField`	Electric Field	Vcm^{-1}
`CurrentDensity`	Total current density	Acm^{-2}
`eCurrentDensity`	Electron current density	Acm^{-2}
`ICurrentDensity`	Iodide current density	Acm^{-2}
`I3CurrentDensity`	Triiodide current density	Acm^{-2}
`CCurrentDensity`	Cation current density	Acm^{-2}

Table 7.3: Elemental quantities. The flag `Current` allows to plot all of them in the x, y and z components.

Scalar quantities

`ContactCurrents`	Contact currents	*[a]

[a] depends on dimension and symmetry

Table 7.4: Scalar quantities.

Part II

Theory

CHAPTER

EIGHT

ELASTICITY

Details about the theory of continuous elasticity applied to device mechanical deformation can be found in [Povolotskiy] .

TiberCAD computes the mechanical deformation of a body subjected to external forces by means of the equilibrium mechanical equation, i.e.

$$\frac{\partial \sigma_{ij}}{\partial x_j} = f_i$$

wheres σ is the stress second-rank tensor and f is the external body force applied to the system. As we will see below, any strain and stress source can be appropriately mapped into an equivalent body force.

The stress tensor σ is related to the mechanical strain by means of the constitutive relationships $\sigma_{ij} = C_{ijlk}\epsilon_{lk}$ where **C** is the stiffness fourth-rank tensor. The strain is related to the displacement u by the expression

$$\epsilon_{kl} = \frac{1}{2}\left(\frac{\partial u_l}{x_k} - \frac{\partial u_k}{\partial u_l}\right)$$

Relying on the symmetry of the strain and stiffness tensor the final equation reads as

$$\frac{\partial}{\partial x_j} C_{ijlk} \frac{\partial u_l}{\partial x_k} = f_i$$

The non-linear strain is computed by applying the mechanical equilibrium equation on the deformed mesh. The number of the shape iteration is set by the keyword **shape_iteration** (default = 0, i.e. no deformation is performed) whereas the maximum tolerated error can be indicated with shape_error (default = $1e^{-3}$).

8.1 Stiffness constant

By default the stiffness constant is anisotropic. For the zincoblend crystal structure the independent coefficients are (with the Voigt notation) C11, C12, C44.

$$C = \begin{pmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{44} \end{pmatrix} \tag{8.1}$$

For the wurtzite structure we have C11, C12, C13 and C44.

$$C = \begin{pmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{13} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{C_{11}-C_{22}}{2} \end{pmatrix} \tag{8.2}$$

An isotropic stiffness can be included with the keyword **Stiffness isotropic** . In this case, the only independent parameters are the Young modulus (Youngin GPa) and the Poisson's ratio (Poisson).

$$C = \frac{E}{(1+\nu)(1-2\nu)} \begin{pmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1-2\nu}{2} \end{pmatrix} \tag{8.3}$$

While the anisotropic model is included by default, the isotropic one must be explicitly indicated

Example:

```
Stiffness isotropic
  {
   Young = 129
   Poisson = 0.349
  }
```

Boundary conditions:

Surfaces forces f^0 are applied by imposing $\sigma_{ij}n_j = f_i^0$ along the surface with normal n. This boundary condition can be used with the keyword **surface_force** . The force can be specified in GPa with force. An example is shown below

```
Contact base
  {
   type = surface_force
   force = (0,0,0.5)
  }
```

On the other hand, one may want to fix some surface of the device. The keyword **clamp** freezes all nodes of a given surface.

Example:

```
Contact substrate
  {
   type = clamp
  }
```

Body forces

A constant value body force can be included by means of the keyword **Bodyforceconstant** .

Example:

```
BodyForceconstant
  {
   force = (0,1,0)
  }
```

When two crystals with different crystal structure are put in contact the lattice mismatch between them may induce a strain $\epsilon^L M$ and, therefore, a stress $\sigma = C\epsilon^{LM}$.

This stress contribution acts as a body force

$$f_i = -\frac{\partial}{\partial x_j} C_{ijlk} \epsilon_{lk}^{LM}$$

The strain source can be computed only once the reference lattice is identified. The reference material and its growth axis can be included following the same syntax of the device section.

Example:

```
BodyForcelattice_mismatch
  {
   reference_material = AlGaN
   structure = wz
   x = 0.2
   x-growth-direction = (1,0,0,0)
   y-growth-direction = (0,1,-1,0)
   z-growth-direction = (0,0,0,1)
  }
```

In presence of an electric field there might develop an additional stress source due to the so-called converse piezoelectric effect, given by :

$$\sigma_{il}^{SC} = -e_{ijk}E_k$$

The converse piezoelectric effect can be included with the keyword **BodyForce_piezo** and an electrostatic simulation must be indicated.

The effective body force reads as :

$$f_i = -\frac{\partial}{\partial x_j}\sigma_{ij}^{CP}$$

Example:

```
BodyForceconverse_piezo
  {
   poisson_simulation = dd
  }
```

Considering all the above mentioned force sources, the plotted total stress and strain are :

$$\sigma_{ij} = C_{ijlk}\left(\frac{\partial u_l}{\partial x_k} + \epsilon_{lk}^{LM}\right) - e_{ijk}E_k$$

$$\epsilon_{kl} = \frac{1}{2}\left(\frac{\partial u_l}{x_k} - \frac{\partial u_k}{\partial u_l}\right) + \epsilon_{lk}^{LM}$$

CHAPTER

NINE

DRIFT-DIFFUSION SIMULATION OF ELECTRONS AND HOLES

9.1 Theory

The semi-classical transport simulation of electrons and holes is based on the drift- diffusion approximation (see [Selberherr]).

Beside the electric potential the electro-chemical potentials are used as variables such that the system of PDEs to be solved reads as follows

$$\begin{aligned} -\nabla(\varepsilon\nabla\varphi - \mathbf{P}) &= -e(n - p - N_d^+ + N_a^-) \\ -\nabla(\mu_n n(\nabla\phi_n + P_n\nabla T)) &= R \\ -\nabla(\mu_p p(\nabla\phi_p + P_p\nabla T)) &= -R \end{aligned} \tag{9.1}$$

P is the electric polarization due to e.g. piezoelectric effects and *R* is the net recombination rate, i.e. recombination rate minus generation rate. P_n and P_p are the electron and hole thermoelectric power, respectively. The models for the mobilities and the net recombination rates can be specified in the **physical_model** sections as described in the following.

9.2 Solution/Plot variables

The solution variables available for plotting and for interaction with other models are given in *Table Plotting variables* .

9.3 Module options

The following options influence the behaviour of the Drift-Diffusion module:

`coupling = string` defines which equations to solve. The default is full, meaning that the full system consisting of the Poisson, electron continuity and hole continuity equations is solved. Other possible values are poisson (for equilibrium calculations), electrons or holes. For the last two cases local equilibrium is assumed such that $\phi_n = \phi_p$.

`enforce_local_charge_neutrality = bool` If set to true, local charge neutrality will be enforced by setting the charge density to zero. This may be useful for solving the Poisson equation involving only dielectrics.

`guess_el_qfermi = double` If this option is set, then before resolving the system the given number will be set as a guess for the electron electro-chemical potential.

`guess_hl_qfermi = double` If this option is set, then before resolving the system the given number will be set as a guess for the hole electro-chemical potential.

`default_boundary_condition = string` With this option the user can control the default boundary condition for the electric field on all external boundaries without explicit boundary model. Possible values are zero field (default), or zero displacement. The two differ only in presence of electric polarization fields.

`quadrature_rule = string` This option allows to chose between trapezoidal and Gauss type numeric integration rules. The default rule is gauss, but in some cases trapez may prevent density peaks near badly resolved material interfaces.

`save_state = boolean` If set to *true* the current solution will be written to a compressed file after each solve. The file name follows the same rules as the result files, having file extension `.tsv` .

`load_state = file` Reload a formerly saved solution. filename can be a filename (to reside in the current working directory), a relative or an absolute path to a file. The file needs to have been created with `save_state = true` .

`solve_after_load = boolean` If set to true, the system will be solved after having reloaded a saved state. Otherwise it will not be solved, which is the default behaviour.

Currently the reload of saved solutions *only works correctly when using the identical mesh* . Otherwise there will be undefined behaviour or failure. Future releases will relax this restriction.

9.4 Solver section

The **Solver** section of the Drift-Diffusion module refers to a nonlinear solver .

9.5 Physics section

The Physics block contains generic options for the bulk physical model and the definition of submodels. The generic options are:

`model = string` Specify the semiconductor model to be used. Possible values are default and simple. The former uses k.p to calculate the (strain corrected) band parame- ters.

`thermal_simulation = string` If you are doing coupled electrothermal simulations, you have to specify the name of the thermal simulation providing the lattice temperature.

`strain_simulation = string` If you are doing simulations on strained systems, you can specify the name of a strain simulation. The band parameters will then be calculated taking local strain corrections into account.

`relax_polarization = double` With this option one can specify a relaxation factor for the electric polarization field. This can be useful if the amount of total electric polarization has to be treated as fitting parameter.

For the simple semiconductor model one has to provide conduction and valence band edges and the effective density of states masses (or the effective density of states itself) in the `Region` sections. The corresponding keywords are given in table 2.2 . In the following we describe the submodels. All submodels can be restricted to a subset of simulation regions.

9.5.1 Recombination/generation models

This section describes the currently available generation/recombination models. Note that all recombination models can be applied also to surfaces/interfaces. Recombination models are controlled by means of recombination submodel blocks inside the Physics section. Different recombination models having the same (or no) options can be enabled in a single statement writing:

```
recombination (model1, model2, ...) {}
```

Shockley-Read-Hall (SRH) recombination

The SRH recombination model can be enabled by defining a recombination submodel of type `srh` .

SRH recombination is defined as follows:

$$R_{SRH} = \frac{np - n_i^2}{(n + n_i e^{E^*/k_B T})\tau_p + (p + n_i e^{-E^*/k_B T})\tau_n} \tag{9.2}$$

$E^* = E_{trap} - (E_c + E_v)/2$ is the trap level with respect to the midband energy.

n_i is the intrinsic carrier density, τ_n and τ_p are the recombination times.

The parameters are taken from the material database. The recombination times are dependent on temperature and doping density, e.g.

$$\tau_n = \tau_n^0 \left(\frac{T}{T_0}\right)^{\alpha_n} e^{\beta(T/T0-1)} \tag{9.3}$$

$$\tau_n^0 = \tau_{min,n} + \frac{\tau_{max,n} - \tau_{min,n}}{1 + (N/N_{ref})^{\gamma}} \tag{9.4}$$

where T_0 is the reference temperature (300 K). Table 2.3 shows the corresponding parameters for the material data files. The parameters for holes and electrons have to be specified in an array, e.g. $\tau_m in = (1e-5, 3e-6)$

The recombination times and trap level can be overridden from the input file by using the keywords of Table 2.4.

The SRH recombination model can be applied also to surfaces and interfaces. In this case, you can provide the recombination velocities using the keywords rec_velocity_n and `rec_velocity_p` instead of `tau_n` and `tau_p` .

Direct (radiative) recombination

The direct recombination model can be enabled in the input file by by defining a `recombination` submodel of type `direct` .

Direct recombination is modeled as follows:

$$R_{direct} = C(np - n_i^2) \tag{9.5}$$

The material data file and the input file use the same keyword C for the parameter C. The database value can be overridden from the input file as described for SRH recombination.

Auger recombination

The Auger recombination model can be enabled in the input file by defining a recombination submodel of type `auger` .

Auger recombination is modeled by the following equation

$$R_{auger} = (C_n n + C_p p)(np - n_i^2) \tag{9.6}$$

with temperature dependent parameters

$$C_{\{n,p\}} = \left(A + B\frac{T}{T_0} + C\left(\frac{T}{T_0}\right)^2\right)\left(1 + He^{-\{n,p\}/N_0}\right)$$

The parameters A;B;C;H and N_0 are taken exclusively from the database. They are different for C_n and C_p and have to be specified as arrays with keywords A, B, C, H, N0, e.g. A = (1e-31, 1e-32). The calculated values for C_n and C_p can be overridden from the input file by specifying values for the keywords C_n and C_p .

Optical generation

A very simple model for photoelectric generation of electron-hole pairs is implemented in TIBERCAD. It is enabled by specifying a `generation` submodel of type optical The model imposes a constant generation rate which has to be provided by the keyword G in units of $(cm * s)^{-1}$.


Note that the simulation usually should define a sweep on the value of G from 0 to the desired generation.

9.5.2 Thermoelectric power models

The thermoelectric power models are the same for electrons and holes. The keyword is `thermoelectric_power` , i.e

```
thermoelectric_power [type]
  {
  }
```

The model keyword can be `constant` (i.e. the thermoelectric powers are read from the database) or `diffusivity_model` where the thermoelectric powers are computed by

$$P_n = -\frac{k_b}{q}\left(\frac{5}{2} + \frac{e\phi_n + E_c - e\varphi}{k_b T}\right) \tag{9.7}$$

$$P_p = \frac{k_b}{q}\left(\frac{5}{2} - \frac{e\phi_p + E_v - e\varphi}{k_b T}\right) \tag{9.8}$$

The default is $P_n = P_p = 0$

9.5.3 Mobility models

The models to be used for electrons and holes can be defined in a single submodel block or independently using two blocks. The corresponding keywords are mobility or

`electron_mobility` and `hole_mobility` , i.e.

```
mobility [type]
  {
  }

electron_mobility [type]
  {
```

```
  }

hole_mobility [type]
  {
  }
```

When using the first approach, both carriers will use the same model, and param- eters provided in the input file will also be used by both carriers. When mixing the difierent definitions, the blocks `electron_mobility` and `hole_mobility` will override the common `mobility` block.

The default model is the constant mobility model. The parameters for the difierent mobility models are needed for both electrons and holes. In the material files they are specified with a common keyword in arrays, e.g.

mobility	constants electrons holes
mu_max	(1400.0 , 250.0)
exponent	(1.0 , 2.1)

9.6 Constant mobility model

The constant mobility model (identifier `constant`) assumes a mobility which depends only on temperature by means of the following formula:

$$\mu_{const} = \mu_0 (T/T_0)^{-\gamma} \tag{9.9}$$

In the material data file μ_0 and γ have to be specified with the keywords mu_max and exponent. μ_0 can be ovverridden from the `physical_model` section using the keyword mu or from the Region sections using the keywords `mu_e` and `mu_h` .

9.7 Doping dependent mobility model

The doping dependent mobility model (identifier `doping_dependent`) implements two models for mobility depending on the total doping density and the temperature. The model that is used depends on the value of the `mobility_formula` parameter.

Model by Masetti et al. [4]

The model by Masetti et al. is identified by `mobility_formula` = 1. It uses the following formula:

$$\mu = \mu_{min,1} * \mathrm{e}^{-P_c/N} + \frac{\mu_{const} - \mu_{min,2}}{1 + (N/C_r)^{\alpha}} - \frac{\mu_1}{1 + (C_s/N)^{\beta}} \tag{9.10}$$

where N is the total doping density and μ_{const} the mobility obtained from the constant mobility model. The parameters are specified in the material file as given in Table 2.5.

Model by Arora [5] The model by Arora is identified by `mobility_formula` = 2. It reads:

$$\mu = \mu_{min} + \frac{\mu_d}{1 + (N/N_0)^{A^*}} \tag{9.11}$$

with

$$\mu_{min} = A_{min}(T/T_0)^{\alpha_m}, \quad \mu_d = A_d(T/T_0)^{\alpha_d}$$
$$N_0 = A_N(T/T_0)^{\alpha_N}, \quad A^* = A_a(T/T_0)^{\alpha_a}$$

The parameters are given in table below.

Field dependent mobility model

The field dependent mobility model describes the degradation of mobility at high driving fields. It is identified by the identifier field_dependent. The electric field component in directon of the current ow or the gradient of the electro-chemical potential can be chosen as driving force:

```
driving_force = efield | grad_fermi | field_parameter
```

The default driving force is the gradient of the corresponding electro-chemical potential $\nabla\phi$. `field_parameter` uses a field parameter given by $\sqrt{E \cdot \nabla\phi}$ as driving force.

The model is based on the Caughey-Thomas model, refined by Canali [6]:

$$\mu = \frac{\mu_{lowfield}}{\left(1 + \left(\frac{\mu_{lowfield}|\mathbf{E}|}{v_{sat}}\right)^{\beta}\right)^{1/\beta}} \tag{9.12}$$

with

$$\beta = \beta_0(T/T_0)^b$$

$|E|$ is the modulus of the driving field, $\mu_{lowfield}$ is the low-field mobility. For the latter one can specify the model to be used using the parameter `lowfield_model` . As default the doping dependent model is used. There are two models for vsat, identified with `Vsat_Formula` = 1 and 2.

Formula 1 reads

$$v_{sat} = v_{sat,0}(T/T_0)^{-\gamma}$$

Formula 2 reads

$$v_{sat} = \max(A_{vsat} - B_{vsat}(T/T_0), v_{min})$$

The parameters for the field dependent mobility model are summarized in Table 2.7.

9.7.1 Polarization models

For simulations involving materials with nonzero electric polarization (such as nitrides) it is important to include the efiect of polarization. This is done by specifying the models for spontaneous (pyro-) and piezoelectric polarization using the keywords polarization with the types `pyro` and `piezo`

```
polarization pyro {}
polarization piezo {}
```

As for all models, if they do not have individual options, they can be specified together by writing **polarization (pyro, piezo) {}** .

Spontaneous (pyro-) polarization

The spontaneous polarization model imposes a constant electric polarization P along the symmetry-braking direction of the crystal. Crystals with wurtzite structure like Ni- trides have strong piezoelectric fields along the c-direction. The value of the polarization usually is taken from the database, but it can be overridden from the input file by speci- fying the option Pz, meaning the value of the spontaneous polarization along c-direction ([0001]). Alternatively, one can specify explicitly a polarization vector using the option

P = (Px, Py, Pz) . This is useful to impose an arbitrary constant polarization field.

Piezopolarization

The piezoelectric polarization is strain induced and given by

$$P^{pz} = e_{ikl}\varepsilon_{kl} \tag{9.13}$$

where ε_{kl} is the strain tensor. The piezoelectric moduli e_{ikl} are stored in the database. The strain is obtained from the simulation specified in the `Physics` section, but it can be overridden by providing a name for the strain simulation inside the polarization block using the `strain_simulation` option.

9.7.2 particle density

Details for the calculation of the electron and hole densities can be given in the particle_density submodel. Its options are:

`particle = string` The particle this model is describing. Can be `electron` or `hole` .

`statistics = string` The statistics. Can be `fermidirac` (default) or `boltzmann` .

`quantum_density = string` **The name of a quantum density simulation. This will use** the quantum mechanical particle density in the regions it was calculated.

If a quantum density is used, then it is useful to define also an embracing region where the model gradually switches from a fully classical to a fully quantum density. The options for the embracing are specified in a block with keyword `embracing` . It accepts the following options:

`embracing_length = double` When the domain of the quantum simulation is smaller than the domain of the full simulation, the boundary conditions for the Schroedinger equation will disturb the transfer from classical to quantum density. By defining an embracing region of a certain extension (specified in meters), a gradual transition from classical to quantum density will be done instead of an abrupt one, using as effective density $\rho = x \cdot \rho_{\text{quantum}} + (1 - x) \cdot \rho_{\text{classical}}$. The default is no embracing region at all (zero extension).

`cutoff = double` if an embracing region is used, a part of this region near the boundary of the quantum region can be cut off so that only the classical density is considered in that part. `cutoff` is specified as a percentage of the embracing length and should therefore be between 0.0 and 1.0.

`plot_embracing_region = bool` Whereas the automatic creation of the embracing region in 1D is a very simple task, it is a more difficult one in higher dimensions. By setting this flag to true, the embracing region and the mixing coefficient x will be plotted for a visual control of the quality of the embracing region. The default is `false` .

9.7.3 Trap models

Currently single level traps are implemented in TiberCAD. Traps can be normally neutral or normally charged electron or hole traps, or a fixed charge. Common options for all models are

`type = string` (If not provided as second keyword). The type of traps. One of eNeutral,

hNeutral, donor, acceptor or fixed_charge.

`Nt = double` The trap density in cm 3 (or cm 2 for surface traps).

`Et = double` The trap level with respect to a reference energy.

reference = string The reference energy. The default is m. The trap energy in this case is given as $E_{trap} = E_{midgap} + Et$. Other possible values are $E_{trap} = E_c - Et$ or $E_{trap} = E_v + Et$.

`eNeutral` The trapped electron density is given by

$$n_t = \frac{N_t}{1 + \exp(\frac{E_{trap} - E_{F,n}}{k_B T})}$$

“hNeutral“ The trapped hole density is given by

$$p_t = \frac{N_t}{1 + \exp(-\frac{E_{trap} - E_{F,p}}{k_B T})}$$

“donor“ The density of ionized traps is given by

$$N_t^+ = N_t - \frac{N_t}{1 + \exp(\frac{E_{trap} - E_{F,n}}{k_B T})}$$

“acceptor“ The density of ionized traps is given by

$$N_t^- = N_t - \frac{N_t}{1 + \exp(-\frac{E_{trap} - E_{F,p}}{k_B T})} \tag{9.14}$$

If traps are specified, the total charge density in the Poisson equation is modified to include the charged trap densities:

$$\rho = e\left(p - n + N_D^+ - N_A^- - \sum n_t + \sum p_t + \sum N_t^+ - \sum N_t^-\right) \tag{9.15}$$

Additionally, each trap induces a SRH recombination term of the form

$$R_t = N_t \frac{v_{th}^n \sigma^n v_{th}^p \sigma^p (np - n_i^2)}{v_{th}^n \sigma^n (n + n_1) + v_{th}^p \sigma^p (p + p1)} \tag{9.16}$$

where $\sigma^{n,p}$ are the capture cross sections, $v_{th}^{n,p}$ the thermal velocities and (for Boltzmann statistics)

$$n_1 = n_{i,eff} \exp(E_{trap}/k_B T), \quad p_1 = n_{i,eff} \exp(-E_{trap}/k_B T) \tag{9.17}$$

9.7.4 Boundary conditions

Boundary conditions are implemented for ohmic contacts, Schottky contacts, free surfaces and interfaces. Contacts are boundary models that allow a nonzero normal electrical current. The applied voltage is specified with the option `voltage` . A variable can be assigned to this, using the $-syntax. On ohmic or schottky contacts one can define surface recombination velocities for

electrons and holes using the options `rec_velocity_e` and `rec_velocity_h` . This will impose Robin type boundary conditions for the continuity equations of the form

$$-\nabla(\mu_n n \nabla\phi_n) = v_n(n - n_0) \quad (9.18)$$

$$-\nabla(\mu_p p(\nabla\phi_p + P_p \nabla T)) = -v_p(p - p0) \quad (9.19)$$

The options `zero_field` , `zero_grad_fermi_e` and `zero_grad_fermi_h` can be used, which when set to `true` will impose zero normal electric field and zero normal gradient of the electron and hole electro-chemical potential, respectively. The latter are special cases of surface recombination velocities ($v_r ec = 0$).

Contacts are defined by blocks with keyword Contact, for example:

```
Contact anode
  {
   type = ohmic
   [regions = (anode1, anode2)]
   voltage = $Vd
  }
```

An area factor can be specified for contacts using the keyword `area_factor` . The contact current will be multiplied by this factor. For interfaces and surfaces, the same syntax can be used (optionally one can use the keywords `Interface` or `Boundary`), however they do usually not need to be defined explicitly.

Ohmic contact

The ohmic contact (identifier ohmic) has no further parameters.

Schottky contact

A Schottky contact (identifier `schottky`) has the additional parameter `barrier` , which signifies the energy difference between the semiconductor band edge and the fermi energy in the metal. As default, the barrier is taken with respect to the conduction band. By specifying `band = v` the barrier can be imposed with respect to the valence band (p- type contact). Alternatively, the metal work function can be defined using the keyword work_function. Note, however, that its value has to be aligned with the band energies given in the material files for the other materials. The `fixed_barrier` controls the behaviour of the barrier height for strained materials. If it is set to true, the barrier will be independent of strain (default behaviour). If it is set to `false` , the given barrier is used as barrier for the unstrained case and will depend on strain during simulation. If the metal work function is specified, the barrier will be strain dependent as default.

 if the contact is touching different materials, one should specify the work function instead of the barrier.

Thermionic emission is by default switched on, but can be disabled by specifying thermionic_emission = false.

Interface/surface model

The free surface or interface model (identifier interface) can include surface charges due to traps and surface recombination. Their definition can be found in section (vedi sopra).

Each trap model will induce automatically a SRH recombination model as in the bulk case.

9.8 Schroedinger/Poisson/Drift-Diffusion calculations

TIBERCAD is able to do selfconsistent Schroedinger-Poisson or Schroedinger-Drift-Diffusion calculations. For this purpose, quantum_density has to be specified for at least one of the carriers, and a selfconsistent simulation should be defined in the Selfconsistent block. The following options { to be specified in the Physics section { control the behaviour of the selfconsistent simulation.

use_density_predictor = bool When set to true, a predictor-corrector scheme will be adopted in the selfconsistent cycle. The Poisson/Drift-Diffusion solver does not just take the particle densities as given by the Schroedinger calculation, but it will assume a dependency of the density on the potentials of the form

$$\rho(\varphi, \phi_n, \phi_p) = \frac{\rho_{\text{quantum}}(\varphi^0, \phi_n^0, \phi_p^0)}{\rho_{\text{classical}}(\varphi^0, \phi_n^0, \phi_p^0)} \rho_{\text{classical}}(\varphi, \phi_n, \phi_p) \tag{9.20}$$

where $(\varphi^0, \phi_n^0, \phi_p^0)$ are the potentials for which the quantum density was calculated. use_density_predictor = true is the preferred method for selfconsistent Schroedinger-Poisson/Drift-Diffusion calculations and is enabled by default.

9.9 Example

The following example shows the Drift-Diffusion module definition for a pn junction.

```
Module driftdiffusion
  {
   name = dd
   #regions = (pside, nside)
   Solver linesearch
     {
     }
   Physics
     {
      recombination srh {}

      mobility doping_dependent {}
```

```
      Contact anode
        {
         voltage = $Vd
        }

        Contact cathode
          {
           voltage = 0
          }
    }
  }
```

Listing 3: Models section for drift-diffusion

Key	Description	Units
`Ec`	Conduction band edge	eV
`Ev`	Valence band edge	eV
`eQFermi`	Electro-chemical potential of electrons	eV ($-e\phi_n$)
`hQFermi`	Electro-chemical potential of holes	eV ($-e\phi_p$)
`Ec0`	Conduction band edge without electric potential	eV
`Ev0`	Valence band edge without electric potential	eV
`Eg`	Band gap	eV
`ConductionBands`	Minima of all conduction bands	eV
`ValenceBands`	Maxima of all valence bands	eV
`ElPotential`	Electric potential	V
`eDensity`	Electron density	cm^{-3}
`hDensity`	Hole density	cm^{-3}
`eMobility`	Electron mobility	$cm^2V^{-1}s^{-1}$
`hMobility`	Hole mobility	$cm^2V^{-1}s^{-1}$
`eConductivity`	Electron conductivity	S/cm
`hConductivity`	Hole conductivity	S/cm
`ElField`	Electric Field	Vcm^{-1}
`Polarization`	Electric Polarization	Cm^{-2}
`CurrentDensity`	Total current density	Acm^{-2}
`eCurrentDensity`	Electron current density	Acm^{-2}
`hCurrentDensity`	Hole current density	Acm^{-2}
`IonizedDonors`	Ionized donor density	cm^{-3}
`IonizedAcceptors`	Ionized acceptor density	cm^{-3}
`IonizedElectronTraps`	Ionized electron trap density	cm^{-3}
`IonizedHoleTraps`	Ionized hole trap density	cm^{-3}
`eThElPower`	Electron thermoelectric power	$V\ K^{-1}$
`hThElPower`	Hole thermoelectric power	$V\ K^{-1}$
`NetRecombination`	The net recombination rate for each recombination/generation model and the total rate	$cm^{-3}s^{-1}$
`ContactCurrent`	The electric current on each contact	A/cm^{3-d}

Table 9.1: Solution/Plot variables

keyword	*description*
`Ec`	conduction band edge (eV)
`Ev`	valence band edge (eV)
`m_dos_e`	conduction band effective DOS mass (m_e)
`m_dos_h`	valence band effective DOS mass (m_e)
`Nc`	conduction band effective DOS (cm^-3)
`Nv`	valence band effective DOS (cm^-3)

Table 9.2: Parameters for the simple semiconductor model

parameter	*keyword*
τ_{min}	`taumin`
τ_{max}	`taumax`
N_{ref}	`Nref`
γ	`gamma`
E^*	`Etrap`
α	`Talpha`
β	`Tcoeff`

Table 9.3: SRH material data file parameters

tau_n	`tau_n`
tau_p	`tau_p`
E^*	`E_t`

Table 9.4: SRH input file parameters

parameter	*keyword*
$\mu_{min,1}$	`mumin1`
$\mu_{min,2}$	`mumin2`
μ_1	`mu1`
P_c	`Pc`
C_r	`Cr`
C_s	`Cs`
α	`alpha`
β	`beta`

Table 9.5: Data file parameters for the mobility model by Masetti et al.

parameter	*keyword*
A_{min}	`mumin`
A_d	`mud`
A_N	`N0`
A_a	`A`
α_m	`am`
α_d	`ad`
α_N	`aN`
α_a	`aA`

Table 9.6: Data file parameters for the mobility model by Arora.

parameter	*keyword*
β_0	`beta0`
b	`betaexp`
$v_{sat,0}$	`vsat0`
γ	`vsatexp`
A_{vsat}	`A_vsat`
B_{vsat}	`B_vsat`
v_{min}	`vsat_min`

Table 9.7: Data file parameters for the mobility model by Arora.

CHAPTER

TEN

THERMAL

Details about irreversible thermodynamics can be found in [Wachutka] .

The heat flux, within the diffusive approach, is given by the Fourier's law, i.e.

$$J_i = -\kappa_{ij} \frac{\partial}{\partial x_j} T \tag{10.1}$$

where κ is thermal conductivity second-rank tensor (which is assumed temperature in dependent). The power balance is ensured by the continuity equation for the thermal

$$\frac{\partial}{\partial x_i} J_i = H \tag{10.2}$$

10.1 Boundary conditions

The boundaries of the thermal simulation domain are set to ideally thermal insulator by default. Dirichlet boundary conditions can be imposed with the keyword boundary *heat_reservoir* .

```
Contact reservoir
  {
   temperature = 300
  }
```

A surface boundary resistance can be added with the keyword *surface_resistance* .

```
Contact surface_resistance
  {
   r_surf = 0.01
   temperature = 300
  }
```

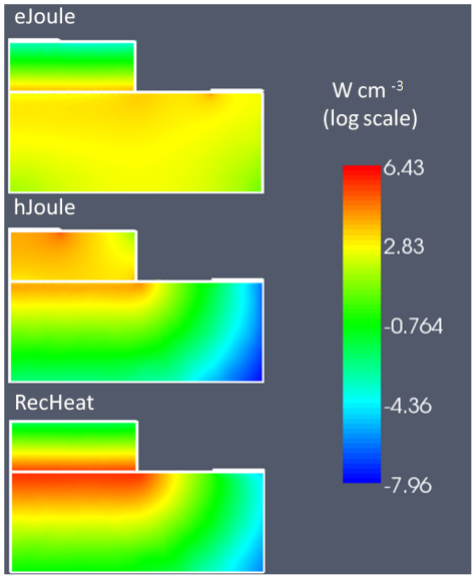
10.2 Heat sources

The constant value heat source can be included with *heat_source constant* .

Example:

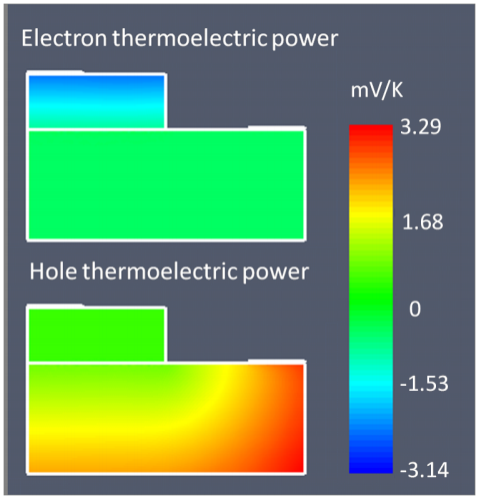
```
HeatSource constant{ H=1e6 }
```

The heat source must be in W/cm^3 . Heat generated by the Joule's effect is included with *heat_source joule* . The name of the drift-diffusion simulation is given by *dd_simulation* .



Example:

```
HeatSource joule{ dd_simulation = dd }
```



CHAPTER

ELEVEN

QUANTUM EFA CALCULATIONS

In TiberCAD, it is possible to perform quantum calculations in the framework of Envelope Function Approximation (EFA): eigenstates and eigenfunctions of a given system, dispersion of quantum states and particle quantum density can be obtained respectively by means of the following Modules:

- Module efaschroedinger
- Module quantumdispersion
- Module quantumdensity

The optical properties are calculated by the following modules

- Module opticskp
- Module opticalspectrum

11.1 Module efaschroedinger

The efaschroedinger simulation tool of TIBERCAD is developed in order to solve a single-particle Schroedinger equation for electrons and holes in a semi- conductor crystal. This problem is an eigenvalue problem that is treated as a generalized complex eigenvalue problem

$$H\psi = ES\psi, \tag{11.1}$$

where H and S are the Hamiltonian and S-matrix, respectively. The EFA calculations are performed by the **Module** efaschroedinger A typical ex- ample is the following:

```
Module efaschroedinger
  {
   particle = el
   poisson_model_name = dd
   strain_model_name = strain # macrostrain
   name = quantum_el
```

```
  regions = quantum
  plot = (EigenFunctions, EigenEnergy, EnergyLevels)
  Solver
    {
     number_of_eigenstates = 10 # 30
    }
  Physics
    {
     model = conduction_band
    }
}
```

11.1.1 Module options

The following options infiuence the behaviour of the Module efaschroedinger:

`particle = string` defines for which particle (electron or hole) Schroedinger equation is

solved Possible values are el and hl. A different Module efaschroedinger has to be defined for each particle to be solved.

`poisson_model_name = string` defines the name of the simulation (Module driftdiffu-

sion) that can provide electric potential

`strain_model_name = string` defines the name of the simulation (Module macrostrain)

that can provide elastic strain

`regions = string` defines the regions associated to this EFA simulation

11.1.2 Solver section

The Solver section of the Module efaschroedinger contains the following options:

`number_of_eigenstates = integer` defines the number of eigenvalues and eigenfunctions to be found.

`Dirichlet_bc_everywhere = boolean` : if true (default value), Dirichlet boundary

conditions are imposed over all the boundaries of the simulation region

`solver = string` : defines the solver for the eigenvalue problem, possible values are:

arnoldi, lapack, krylovshur. The default value is **krylovshur** .

In the case of the lapack solver all the eigenvalues are computed. In the case of arnoldi or krylovshur solver it is necessary to specify which and how many eigenvalues have to be computed. The idea is that the iterative solver calculates several eigenvalues that are close to a specific number, refered to as the *spectral_shift*

`max_iteration_number = integer` : maximum number of iteration, used as a stop condition

`eigen_solver_tolerance = double` :numerical eigensolver tolerance used as a convergence criteria

`spectral_shift = double` :the closest eigenvalues to this value (eV) are found. If not

defined, then it will be calculated internally from the band edges.

`ksp_type = string` : Krylov subspace method type; bcgsl, gmres, cg

`pc_type = string` : preconditioner type: cholesky, jacobi, ilu , composite.

11.1.3 Physics section

`model = string` : possible values are : conduction band , for single conduction band

model (Γ point) ; kp for $\mathbf{k} \cdot \mathbf{p}$ model

`kp_model = string` : possible values are : 6?6 , 8?8.

11.1.4 Output

The available output variables, to be specified in the plot option, are the following:

- EigenEnergy Eigen energy in eV
- EigenFunctions $|\psi(\mathbf{r})|^2$ function of the eigenstate
- Occupation probability to find the state occupied. It is calculated assuming Fermi

distribution and mean electrochemical potential and temperature:

$$\bar{\mu} = \langle\psi|\mu(\mathbf{r})|\psi\rangle \bar{T} = \langle\psi|T(\mathbf{r})|\psi\rangle \tag{11.2}$$

- EnergyLevels graphical output used for showing the energy level over the band

diagram.

11.2 Module quantumdispersion

With the Module quantumdispersion it is possible to calculate the dependence of quan- tum eigenstates on **k** -vector. Such dependence gives the *quantum state dispersion* . The simulation name is **quantumdispersion** .

```
Module quantumdispersion
  {
   simulation_name = dispersion1D_el
   regions = all
   quantum_simulation = quantum_el
   min_eigenvalue_number = 0
   max_eigenvalue_number = 5
   wedge = half
   k_space_dimension = 1
   k1 = (0, 0.1, 0)
   number_of_nodes = (10)
   output_format = grace
   plot = k-space_dispersion
  }
```

The dispersion of quantum states is calculated at k-points that are nodes of the mesh in k-space.

The main parameters are:

- **quantum simulation** : name of the efaschroedinger simulation.
- **min eigenvalue number** , **max eigenvalue number** : the dispersion is calculated

for the states number *i* , where

$$\text{max\_eigenvalue\_number} \geq i \geq \text{min\_eigenvalue\_number}$$

The rest of the parameters (wedge, k space dimension, etc...) define the k-space.

11.2.1 Output

The output variable name is **k-space_dispersion** . The output format for the dispersion can be controlled independently of the general specification in the `Simulation` section by redefining the `output_format` keyword.

11.3 Module quantumdensity

In Module quantumdensity you can define the calculation of particle (electron,hole) den- sity, based on the result of a previous calculation of the system eigenstates (e.g. with efaschroedinger module). Quantum density may be obtained with an analytical or a numerical calculation.

Analytical calculation of density is done in the following way. For each eigenstate we calculate the effective mass assuming quadratic dispersion. Then the charge density is calculated as follows:

$$\rho_{1D}(\mathbf{r}) = g\frac{mkT}{2\pi\hbar^2}|\psi(\mathbf{r})|^2 \ln\left(1+\exp\left(\pm\frac{\mu-E}{kT}\right)\right) \rho_{2D}(\mathbf{r}) = g|\psi(\mathbf{r})|^2\frac{1}{2}\sqrt{\left(\frac{mkT}{2\pi\hbar^2}\right)}F_{-1/2}\left(\pm\frac{\mu-E}{kT}\right), \tag{11.3}$$

where ρ_{1D} and ρ_{2D} are the 1D and 2D charge densities; m is the averaged mass (the mass is different for each quantized state and is position independent); g is the degeneracy of the states. The + sign is for electrons, the - sign is for holes.

Numerical calculation is done by the following formula:

The integration is performed on a mesh in the k-space.

```
Module quantumdensity
  {
   name = dens_el
   regions = quantum
   plot = quantum_density
   k-space_dimension = 0
   k-space_basis = true
   k1 = (0, 0, 0.1)
   #number_of_nodes = (4)
   #wedge = half
   quantum_simulation = quantum_el
   degeneracy = 2
   refine_fraction = 0.20
   relative_accuracy = 0.01
   refine_k_space = true
   output_density_in_k_space = true
   uniform_refinement = false
   mesh_order = FIRST
   first_state = 3
   initial_eigenstates_number = 10
   analytic = true
  }
```

The available options are:

- **quantum_simulation** : name of the efaschroedinger simulation.
- degeneracy: degeneracy of the quantum state
- **initial_eigenstates_number** : initial number of eigenstates for the Schroedinger

equation

- **analytic** = { true | false } If true then the density is calculated analytically,

otherwise numerically.

11.3.1 Output

The output parameter is **quantum_density**.

11.4 Module opticskp

By defining the Module **opticskp** , calculation of optical properties is enabled; in particular, the optical kp matrix elements are calculated from the quantum models specified in the Module.

The optical spectrum from spontaneous emission is calculated in the following way:

where f_i and f_j are the Fermi distributions.

```
Module opticskp
  {
   name = optics
   regions = quantum
   plot = (optical_spectrum_k_0 )
   initial_state_model = quantum_el
   final_state_model = quantum_hl
   initial_eigenstates = (0, 9)
   final_eigenstates = (0, 15)
   polarization = (0, 0, 1)
   Emin = 2.8
   Emax = 3.6
   dE = 0.001
  }
```

Here, *initial_state_model* and *final_state_model* are, respectively, the quantum simulations (**efaschroedinger** module) associated respectively to the initial state of optical transition (e.g. electron), and to the final state of optical transition (e.g. hole). *initial_eigenstates* and *final_eigenstates* refer to the range of eigenstates to be taken in account for optical calculations.

By specifying a range of energy values in this way:

```
Emin = 3.0
Emax = 5.0
dE = 0.001
```

the emission optical spectrum for **k=0** is calculated.

11.4.1 Output

The output variables for optics calculations are:

- **optical_spectrum_k_0** : optical emission spectrum for *k=0*.

11.5 Module opticalspectrum

By defining the Module **opticalspectrum** , optical matrix elements are used to calculate the associated (emission) spectrum with a k-space integration.

```
Module opticalspectrum
  {
   k_space_dimension = 2
   k-space_basis = true
   k1 = (0, 0, 0.1)
   k2 = (0, 0.1, 0)
   refine_fraction = 0.30
   relative_accuracy = 0.01
   refine_k_space = true
   number_of_nodes = (2, 2)
   wedge = quarter
   plot = (optical_spectrum)
   optical_matr_elem_model = opticskp
   polarization = (0, 0, 1)
   Emin = 3.0
   Emax = 5.0
   dE = 0.001
  }
```

The parameters are the following:

`k_space_dimension` = **1** for 2D simulations, **2** for 1D simulations. k-space basis is

true if the k-space is defined by means of k-vectors; if **false** , vectors are expressed in real space

If `refine_k_space` = **true** , that is adaptive k-mesh refinement is enabled, all the el-

ements whose error is greater than the value **(1-refine fraction)** (maximum error) are going to be refined. In this case, "Error" is just the integrated quantity. The refinement will end when the *relative_accuracy* is obtained.

`number_of_nodes` = numb. of elements in k mesh, along each direction

`wedge` = half | quarter, to reduce calculation time, by exploiting symmetry.

`optical_matr_elem_model` = name of the *opticskp* model associated

`polarization` = light polarization (vector)

`Emin, Emax, dE` : energy range and step of spectrum calculation.

11.5.1 Output

The output variables for optics calculations are:

- **optical_spectrum** : k-space integrated optical emission spectrum.

CHAPTER

TWELVE

SIMULATION DYE SOLAR CELLS

For a brief list of literature to understand Dye Solar cells (DSC) see [Kalyanasundaram] . For a review of the model see [Gagliardi] .

The model consists in a set of drift-diffusion equations for the description of the propagation of different ions and for the electrons coupled with Poisson equation:

$$\begin{aligned} \nabla \cdot (\mu_e n_e \nabla \phi_e) &= (G - R) && (12.1)\\ \nabla \cdot (\mu_{I^-} n_{I^-} \nabla \phi_{I^-}) &= -\frac{3}{2}(G - R) && (12.2)\\ \nabla \cdot (\mu_{I_3^-} n_{I_3^-} \nabla \phi_{I_3^-}) &= \frac{1}{2}(G - R) && (12.3)\\ \nabla \cdot (\mu_c n_c \nabla \phi_c) &= 0, && (12.4) \end{aligned}$$

where μ_α refers to carrier mobilities, n_α to concentrations and ϕ_α to electrochemical potentials. *R* is the recombination term and *G* the generation term due to illumination. The Poisson equation to handle the internal potential drop:

$$-\varepsilon \triangle \varphi = q[n_c + N_D^+ - n_{I^-} - n_{I_3^-} - (n_e - \bar{n}_e)], \tag{12.5}$$

where N_D^+ is the amount of ionized dyes and it is equal to:

$$N_D^+ = \frac{G}{k_3} \tag{12.6}$$

with *G* the generation term and :math‘k_3‘ the rate constant of dye regeneration. The dielectric constant, ϵ , of the mesoporous material is a mix the two dielectric functions of the semi- conductor and the electrolyte. We use the Maxwell-Garnet model where the dielectric function of the mixed medium becomes:

$$\varepsilon = \varepsilon_s \frac{\varepsilon_e + 2\varepsilon_s + 2\epsilon_p \varepsilon_e - 2\varepsilon_s \epsilon_p}{\varepsilon_e + 2\varepsilon_s - \epsilon_p \varepsilon_e + \varepsilon_s \epsilon_p} \tag{12.7}$$

where ϵ_s and ϵ_e are the dielectric constants of the semiconductor and the electrolyte, respectively, and ϵ_p is the porosity of the medium. The recombination term depends largely on the loss mechanisms at the electrolyte/oxide interface which follows the reaction chain:

$$I^- \rightleftharpoons I + e \tag{12.8}$$
$$2I \rightleftharpoons I_2 \tag{12.9}$$
$$I_2 + I^- \rightleftharpoons I_3^-. \tag{12.10}$$

From the chemical path we can get a formula for the interface recombination (considering that the first chemical reaction is the slow process):

$$R = k_e \left[\left(\frac{n_e}{\bar{n}_e} \right)^{\beta} \bar{n}_e \sqrt{\frac{n_{I_3^-}}{n_{I^-}}} - \bar{n}_e \sqrt{\frac{\bar{n}_{I_3^-}}{\bar{n}_{I^-}^3}} n_{I^-} \right], \tag{12.11}$$

where the electron rate k_e is the recombination rate constant.

For the boundary conditions of the model we assume at the photoanode:

- $\phi_e = V$: electrochemical potential of electrons set with the voltage applied;
- $\nabla \phi_{I^-} = 0$: no iodide current at the photoanode;
- $\nabla \phi_{I_3^-} = 0$: no triiodide current at the photoanode;
- $\nabla \phi_c = 0$: no cationic current;
- $\nabla \varphi = 0$: no charged layer at the photoanode;

at the cathode:

- $\nabla \phi_e = 0$: no electronic current at the cathode;
- $-q\mu_{I^-} n_{I^-} \nabla \phi_{I^-} = \frac{3}{2} \left(\frac{-E_{red}(\mathbf{r_c})}{R_L} \right)$: split of the current between the ionic species;
- $-q\mu_{I_3^-} n_{I_3^-} \nabla \phi_{I_3^-} = -\frac{1}{2} \left(\frac{-E_{red}(\mathbf{r}_c)}{R_L} \right)$: split of the current between the ionic species;
- $\nabla \phi_c = 0$: no cationic current;

integral boundary conditions for conservation of ionic species:

- $\int_\Omega \left[\frac{1}{3} n_{I^-}(\mathbf{r}) + n_{I_3^-}(\mathbf{r}) \right] d\mathbf{r} = \left(\frac{1}{3} \bar{n}_{I^-} + \bar{n}_{I_3^-} \right) \Omega$: conservation of iodine ions within the cell;
- $\int_\Omega n_c(\mathbf{r}) d\mathbf{r} = \bar{n}_c \Omega$: conservation of cation within the cell;

where ω is the volume of the cell, $n\alpha$ the density of charged species and the index α stands for cation (c), iodide (I^-), triiodide (I_3^-) and electrons (e). R_L is the external load. The bias applied is equal to:

$$V = \phi_e - E_{red}. \tag{12.12}$$

E_{red} is the redox potential. The redox potential can be evaluated using a Nernst approximation:

$$E_{red} = E^0_{Pt} - \frac{kT}{2} ln \left(\frac{n_{I_3^-}/n_{St}}{(n_{I^-}/n_{St})^3} \right). \tag{12.13}$$

12.1 Module DSC

The DSC module is tagged as dssc. In this part of the input file are set the name of the simulation and the list of plotted variables:

```
Module dssc
  {
   name = dssc
   plot = (Potential, Density, Current, ContactCurrents)

   Solver linesearch
     {
      max_iterations = 300
      step_tolerance = 1e-4

      linear_solver
        {
         preconditioner = lu
        }
     }

   Physics
     {
      ...
     }

   Contact anode
     {
      ...
     }

   Contact cathode
     {
      ...
     }
 }
```

This section contains information about the **Contacts** , parameters for the **Solver** and the **Physics** sections.

12.1.1 Contacts

Information for the contacts is inserted in the **Module** section, in the subsections **Contacts** . The two contacts are the photoanode and the cathode. The photoanode must be the boundary of a region where TiO_2 is present, on the contrary the cathode must be the boundary of a region where the **electrolyte** is present.

```
Contact anode
  {
   type = ohmic
   bias = $V[0.0]
  }

Contact cathode
  {
   type = Pt
   load = $R[1e8]
  }
```

The anode is an ohmic contact, it contains only the sweep of the bias applied for the electrochemical potential of the electrons. The cathode must contain the initial external resistance (`load = R`). If a simulation of the cell under illumination is made the initial value of load must be set to a high value, in the range of $10^6 - 10^8 \Omega cm^2$ in order to have no current during the light intensity sweep. In case a simulation under dark conditions is performed, where an external bias only is applied, load = 1.0.

12.1.2 Physics

For the generation:

```
generation = dssc_generation
```

The **Physics** section must contain at least the setting of the generation module. The set of parameters that can be defined for the entire device are enlisted in table *below* .

12.2 Generation module

The generation term is related to the flux of photons which reaches the active TiO_2 regions and the dye present in the cell. We assume a simple Beer-Lambert exponential decay for charge generation of the form:

$$G(\mathbf{r}) = \int_{\lambda_1}^{\lambda_2} \alpha(\lambda)\Phi(\lambda)e^{-\alpha(\lambda)\hat{n}\cdot\mathbf{r}}d\lambda \tag{12.14}$$

Parameters		
	Poisson equation	
`porosity`	Porosity of porous material	
`perm_oxide`	Relative perm. of the TiO_2	
`perm_electrolyte`	Relative perm. of the electrolyte	
`k_3`	Oxidized Dye regeneration rate constant	s^{-1}
	Drift Diffusion equation	
`ne`	Dark electron density	cm^{-3}
`nI`	Dark iodide density	mol/l
`nI3`	Dark electron density	mol/l
`mu_e`	Electron mobility	cm^2/Vs
`D_I`	Iodide diffusion constant	cm^2/s
`D_I3`	Triiodide diffusion constant	cm^2/s
`D_C`	Cation diffusion constant	cm^2/s
	Recombination term	
`k_e`	Recombination constant rate	s^{-1}
`beta`	Non-linearity exponent in the electron density recombination	

Table 12.1: Physical parameters that can be set for the model divided in subsets relative to different processes in the cell.

where α is the absorption coefficient (in μm^{-1}) of the chosen Dye, $\Phi(\lambda)$ the intensity of the light power at wavelength λ of the light source spectrum.

The part of the input file for the generation module:

```
Module dssc_generation
  {
   regions = TiO2
   plot = (Distance, Generation)
   light_direction = (1, 0, 0)
   light_intensity = $x
   dye = N719

   Contact anode
     {
     }
  }
```

In the generation module must be specified:

- (`regions`) the regions where we want that there is generation (where the Dye is present);
- (`plot`) if we want to plot the generation;
- (`light_direction`) the vector which fixes the direction from where the light comes;

- (`light_intensity`) the light intensity;
- (`dye`) the dye used in the cell;
- (`illumination_spectrum`) the source spectrum, by default a 1.5 AM solar spectrum;

The light intensity is defined in a sweep. It can be set to reach 1 that means one Sun, or a larger or smaller illumination intensity (0.1, 2.0, etc.).

There is another flag called **illumination_spectrum** that can be used if it is wanted to change the spectrum profile F with another illumination source (for example if it is assumed that the cell is under light concentration). The file used by default for Φ is the standard 1.5 AM sun spectrum contained in the material database in the file `Sun1p5am` . The last part of the generation is the definition of the Contact. This Contact tells to the code which is the boundary region of the grid from where the light penetrates into the device. The information given by the boundary and the light direction vector defines the coming of light and direction of rays.

12.3 Device

Device section for a DSC simulation:

```
# Description of the device physical regions Device
  {
   meshfile = <name_of_the_mesh>

   Region TiO2
     {
      material = TiO2mes
      porosity = 0.5
     }

   Region electrolyte
     {
      material = TiO2mes
      TiO2 = false
     }
  }
```

The Device section must contain the name of the mesh file

`meshfile = <name_of_the_mesh>.`

For every region of the device it is specified the material file from the database (the **TiO2mes** contains standard parameters for both TiO2 and electrolyte, so it can be used for both regions).

For every region must be specified if it contains TiO2, or electrolyte or both. This can be done setting two flags called `TiO2` and `electrolyte` . If set **true** the material is present in the region, if set **false** it is not present. By default they are assumed both **true** (porous region). In the

second region of the example showed here there is electrolyte only and **TiO2 = false** is explicitly specified. In case both materials are present (porous region) a porosity must be specified (in the range between 0, TiO2 only, and 1, electrolyte only). If one material is not present the porosity is automatically set to 0 or 1.

12.4 Sweep

Three sweeps are needed for the plot of the entire I-V under illumination. The first sweep is needed to make the transition from dark condition to full open-circuit condition under illumination. The high value of the load R maintains the current to zero. Then a second sweep is used to pass from open circuit condition to short circuit condition lowering the external load from a high external load to a small one (`R = 1`).

Finally, the voltage sweep compute the I-V characteristics under illumination. In case of dark simulation (application of an external bias without illumination) the first and second sweep are not needed. The external load for the cathode can be set directly to `R = 1` .

```
Module sweep
  {
   name = sweep_gen
   solve = (dssc_generation, dssc)
   variable = $x
   values = (0, 1e-9, 1e-8, 1e-7, 1e-6,
             1e-5, 1e-4, 1e-3, 1e-2, 0.1, 1)

   plot_data = true
  }

Module sweep
  {
   name = sweep_R
   solve = dssc
   variable = $R
   values = ( 1000, 100, 10, 1 )
   plot_data = true
  }

Module sweep
  {
   name = sweep_V
   solve = dssc
   variable = $V
   values = (0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.62, 0.64, 0.66,
             0.68, 0.7, 0.72, 0.74, 0.76, 0.78, 0.8)
   plot_data = true
```

```
}
```

The intensity of illumination can be changed sweeping the value of x different to 1 (1 = Sun of power).

12.5 Output

The output that want to be plotted are enlisted in section `Module dssc`.

See tables:

Nodal quantities

	`Potential`	
`eQFermi`	Electron electrochemical potential	eV ($-e\phi_n$)
`IQFermi`	Iodide electrochemical potential	eV ($-e\phi_{I^-}$)
`I3QFermi`	Triiodide electrochemical potential	eV ($-e\phi_{I_3^-}$)
`CQFermi`	Cation electrochemical potential	eV ($-e\phi_C$)
`ElPotential`	Electrostatic potential	eV
`Eredox`	Electrolyte electrochemical potential	eV ($-e\phi_{Red}$)
	`Density`	
`eDensity`	Electron density	cm^{-3}
`IDensity`	Iodide density	cm^{-3}
`I3Density`	Triiodide density	cm^{-3}
`CDensity`	Cation density	cm^{-3}
`Generation`	The net electron generation rate	$cm^{-3}s^{-1}$
`NetRecombination`	The net recombination rate	$cm^{-3}s^{-1}$

Table 12.2: Nodal quantities. The flags `Potential` and `Density` allow to plot all the electrochemical potentials and all the densities, including generation and recombination.

Elemental quantities

	`Current`	
`ElField`	Electric Field	Vcm^{-1}
`CurrentDensity`	Total current density	Acm^{-2}
`eCurrentDensity`	Electron current density	Acm^{-2}
`ICurrentDensity`	Iodide current density	Acm^{-2}
`I3CurrentDensity`	Triiodide current density	Acm^{-2}
`CCurrentDensity`	Cation current density	Acm^{-2}

Table 12.3: Elemental quantities. The flag `Current` allows to plot all of them in the x, y and z components.

Scalar quantities		
`ContactCurrents`	Contact currents	*[a]

[a] depends on dimension and symmetry

Table 12.4: Scalar quantities.

Part III

Reference Guide

CHAPTER

THIRTEEN

ELASTICITY

Stiffness/Isotropic
Kind: bulk
Name: Stiffness
Type: isotropic
database_section: none
Options:
young (double,0.0)
poisson (double [-0.5,1.0],0.0)

Stiffness/Anisotropic
Kind: bulk
Name: Stiffness
Type: anisotropic
database_section = stiffness
Options:
none

Body Force/constant
Kind: bulk
Name: BodyForce
Type: constant
database_section: none
Options:
force (double vector,(0.0,0.0,0.0))

Body Force/Lattice Mismatch
Kind: bulk
Name: BodyForce
Type: lattice_mismatch
database_section: none
Options:
x (double [0.0,1,0],0.0)
x-growth-direction (double vector)
y-growth-direction (double vector)
z-growth-direction (double vector)

Body Force/Converse
Kind: bulk
Name: BodyForce
Type: converse
database_section: piezoelectricity
Options:
poisson_simulation (string,"none")

Boundary/clamp
Kind: interface
Name: Contact
Type: converse
database_section: none
Options:
poisson_simulation (string,"none")

Boundary/Surface Force
Kind: interface
Name: Contact
Type: surface_force
database_section: none
Options:

force (double vector,(0.0,0.0,0.0)

CHAPTER

FOURTEEN

SOLVERS

14.1 Numerical Solvers

the "Solver" block inside a module description contains the options for the numerical solver. The solver type the "Solver" block is describing (linear, nonlinear, eigenvalue solver) depends on the module. For nonlinear and linear solvers, the following options exist:

14.1.1 Nonlinear solvers:

type :

petsc = uses the PETSc nonlinear solver (SNES) (default)

linesearch = uses a linear linesearch implemented in TiberCAD

Nonlinear solvers are based on iterative methods, solving in each iteration a linear system. The linear solver used for this can be controlled by providing a block with keyword "linear_solver" containing the options for the linear solver (see linear solvers).

petsc

- `relative_tolerance` = convergence criterion based on relative residual l_2-norm
- `absolute_tolerance` = convergence criterion based on the l_2-norm of the residual
- `max_iterations` = maximum number of iterations
- `step_tolerance` = tolerance criterion based on the l_2-norm of the correction step
- `max_step` = maximum linesearch step (l_2-norm)
- ‘‘divergence_tolerance = divergence criterion

linesearch

- `absolute_tolerance` = convergence criterion based on the l_2-norm of the residual
- `step_tolerance` = tolerance criterion based on the linfinity-norm of the correction step

- `max_iterations` = maximum number of iterations

14.1.2 Linear solvers

type = petsc

Petsc

`method`, The Krylov subspace method to be used:

`bcgs` = BiCGSTAB `bcgsl` = `gmres` = Generalized Minimal Residual `bicg` = Bi-Conjugate Gradient `cg` = Conjugate Gradient `cgs` = Conjugate Gradient Squared `richardson` = Richardson `pconly` = only apply preconditioner

- `relative_tolerance` = convergence criterion based on relative residual l_2-norm
- `absolute_tolerance` = convergence criterion based on the l_2-norm of the residual
- `max_iterations` = maximum number of iterations

Preconditioner :

`lu` = LU `ilu` = incomplete LU `jacobi` = Jacobi `cholesky` = Cholesky `none` = no preconditioning `composite` = use a combination of Jacobi and iLU

14.2 Simulations

Sweep

Performs a linear sweep for a given variable.

Options:

`solve` = a list of simulations to be solved, eg. (strain, dd)

`variable` = the sweep variable, eg. $Vd

`start` = the first value of the sweep

`stop` = the last value of the sweep

`steps` = the number of steps the interval (start, stop) gets subdivided in

`values` = instead of start, stop and steps, provide the sweep values explicitly

`plot_data` = set to true if you want to plot after each step the simulation results (default is false)

`min_step` = the minimum absolute step size

`max_step` = the maximum absolute step size

`initial_step` = the absolute initial step size

`min_relative_step` = minimum relative step size, default is 1e-3

`max_relative_step` = maximum relative step size, default is 1

`initial_relative_step` = initial relative step size, default is 1

The relative step sizes refer to each single sweep step. If max_relative_step is less than one, each sweep step will be subdivided in smaller steps. If a simulation fails, the step gets reduced by half until the simulation succeeds, or until the minimum relative or absolute step size is reached. In the latter case, the sweep is assumed to have failed. As for any module, a name can be given to a sweep block. This is important when several sweeps are defined, and in particular when nested sweep (solve option of one sweep refers to another sweep) are used.

Example:

```
Module sweep
  {
   solve = (dd, thermal)
   start = 0
   stop = 1
   steps = 10
   plot_data = true
  }
```

14.3 Selfconsistent

Solves models in an iterative way to obtain a selfconsistent solution, optionally using a relaxation approach. options:

- `max_iteration` = the maximum number of iterations. Currently, when the maximum number of iterations is reached, the program only issues a warning and proceeds.
- `relative_tolerance` = the relative convergence tolerance for the observed variable in terms of the l2-norm
- `absolute_tolerance` = the absolute convergence tolerance for the observed variable in terms of the maximum-norm
- `relaxation_factor` = an optional relaxation factor (default is 1) to be applied to the observed variable
- `solve` = the simulations to be solved

Currently, the observed variable on which convergence control and relaxation is done is the system variable of the last simulation specified in 'solve'.

CHAPTER

FIFTEEN

THERMAL

Heat Source/constant
Kind: bulk
Name: HeatSource
Type: constant
database_section: none
Options:
H (double,0.0)

Heat Source/joule
Kind: bulk
Name: HeatSource
Type: joule
database_section: none
Options:
dd_simulation (string,”driftdiffusion”)

Boundary/Heat reservoir
Kind: interface
Name: Contact
Type: heat_reservoir
database_section: none
Options:
temperature (positive double,300)

Boundary/Surface Thermal Resistance
Kind: interface
Name: Contact
Type: surface_resistance
database_section: none
Options:
r_surf (positive double 0.0)
temperature (positive double (K),300)

Boundary/Flux
Kind: interface
Name: Contact
Type: flux
database_section: none
Options:
Flux (double vector (W/cm2, (0.0,0.0,0.0))

CHAPTER

SIXTEEN

QUANTUM EFA CALCULATIONS

In TiberCAD, it is possible to perform quantum calculations in the framework of Envelope Function Approximation (EFA): eigenstates and eigenfunctions of a given system, dispersion of quantum states and particle quantum density can be obtained respectively by means of the following Modules:

- Module efaschroedinger
- Module quantumdispersion
- Module quantumdensity

The optical properties are calculated by the following modules

- Module opticskp
- Module opticalspectrum

16.1 Module efaschroedinger

The EFA calculation of eigenstates and eigenfunctions are performed by the **Module** efaschroedinger. A typical example is the following:

```
Module efaschroedinger
  {
   particle = el
   poisson_model_name = dd
   strain_model_name = strain # macrostrain
   name = quantum_el
   regions = quantum
   plot = (EigenFunctions, EigenEnergy, EnergyLevels)
   Solver
     {
      number_of_eigenstates = 10 # 30
     }
```

```
  Physics
    {
     model = conduction_band
    }
}
```

16.1.1 Module options

The following options influence the behaviour of the Module efaschroedinger:

`particle = string` defines for which particle (electron or hole) Schroedinger equation is

solved Possible values are el and hl. A different Module efaschroedinger has to be defined for each particle to be solved.

`poisson_model_name = string` defines the name of the simulation (Module driftdiffu-

sion) that can provide electric potential

`strain_model_name = string` defines the name of the simulation (Module macrostrain)

that can provide elastic strain

`regions = string` defines the regions associated to this EFA simulation

16.1.2 Solver section

The Solver section of the Module efaschroedinger contains the following options:

`number_of_eigenstates = integer` defines the number of eigenvalues and eigenfunctions to be found.

`Dirichlet_bc_everywhere = boolean` : if true (default value), Dirichlet boundary

conditions are imposed over all the boundaries of the simulation region

`solver = string` : defines the solver for the eigenvalue problem, possible values are:

arnoldi, lapack, krylovshur. The default value is **krylovshur** . In the case of the lapack solver all the eigenvalues are computed. In the case of arnoldi or krylovshur solver it is necessary to specify which and how many eigenvalues have to be computed. The idea is that the iterative solver calculates several eigenvalues that are close to a specific number, refered to as the *spectral_shift*

`max_iteration_number = integer` : maximum number of iteration, used as a stop condition

`eigen_solver_tolerance = double` :numerical eigensolver tolerance used as a convergence criteria

`spectral_shift = double` :the closest eigenvalues to this value (eV) are found. If not

defined, then it will be calculated internally from the band edges.

`ksp_type = string` : Krylov subspace method type; bcgsl, gmres, cg

`pc_type = string` : preconditioner type: cholesky, jacobi, ilu , composite.

16.1.3 Physics section

`model = string` : possible values are : conduction band , for single conduction band

model (Γ point) ; kp for $\mathbf{k} \cdot \mathbf{p}$ model

`kp_model = string` : possible values are : 6?6 , 8?8.

16.1.4 Output

The available output variables, to be specified in the plot option, are the following:

- EigenEnergy Eigen energy in eV
- EigenFunctions $|\psi(\mathbf{r})|^2$ function of the eigenstate
- Occupation probability to find the state occupied. It is calculated assuming Fermi

distribution and mean electrochemical potential and temperature:

- EnergyLevels graphical output used for showing the energy level over the band

diagram.

<<<<<<< .mine By defining the Module **opticskp** , calculation of optical properties is enabled; in particular, the optical kp matrix elements are calculated from the quantum models specified in the Module.

The optical spectrum from spontaneous emission is calculated in the following way:

where f_i and f_j are the Fermi distributions.

```
Module opticskp
  {
   name = optics
```

```
  regions = quantum
  plot = (optical_spectrum_k_0 )
  initial_state_model = quantum_el
  final_state_model = quantum_hl
  initial_eigenstates = (0, 9)
  final_eigenstates = (0, 15)
  polarization = (0, 0, 1)
  Emin = 2.8
  Emax = 3.6
  dE = 0.001
 }
```

Here, *initial_state_model* and *final_state_model* are, respectively, the quantum simulations (**efaschroedinger** module) associated respectively to the initial state of optical transition (e.g. electron), and to the final state of optical transition (e.g. hole). *initial_eigenstates* and *final_eigenstates* refer to the range of eigenstates to be taken in account for optical calculations.

By specifying a range of energy values in this way:

```
Emin = 3.0
Emax = 5.0
dE = 0.001
```

the emission optical spectrum for **k=0** is calculated.

16.2 Output

The output variables for optics calculations are:

- **optical_spectrum_k_0** : optical emission spectrum for *k=0*.

CHAPTER

SEVENTEEN

MODULE OPTICALSPECTRUM

By defining the Module **opticalspectrum** , optical matrix elements are used to calculate the associated (emission) spectrum with a k-space integration.

```
Module opticalspectrum
  {
   k_space_dimension = 2
   k-space_basis = true
   k1 = (0, 0, 0.1)
   k2 = (0, 0.1, 0)
   refine_fraction = 0.30
   relative_accuracy = 0.01
   refine_k_space = true
   number_of_nodes = (2, 2)
   wedge = quarter
   plot = (optical_spectrum)
   optical_matr_elem_model = opticskp
   polarization = (0, 0, 1)
   Emin = 3.0
   Emax = 5.0
   dE = 0.001
  }
```

The parameters are the following:

`k_space_dimension` = **1** for 2D simulations, **2** for 1D simulations. k-space basis is

true if the k-space is defined by means of k-vectors; if **false** , vectors are expressed in

real space

If `refine_k_space` = **true** , that is adaptive k-mesh refinement is enabled, all the el-

ements whose error is greater than the value (1-refine fraction)* (maximum error) are going to be refined. In this case, “Error” is just the integrated quantity. The refinement will end when the

relative_accuracy is obtained.

`number_of_nodes` = numb. of elements in k mesh, along each direction

`wedge` = half | quarter, to reduce calculation time, by exploiting symmetry.

`optical_matr_elem_model` = name of the *opticskp* model associated

`polarization` = light polarization (vector)

`Emin, Emax, dE` : energy range and step of spectrum calculation.

CHAPTER

EIGHTEEN

OUTPUT

Module quantumdispersion >>>>>>> .r2404 ————————

With the Module quantumdispersion it is possible to calculate the dependence of quan- tum eigenstates on **k** -vector. Such dependence gives the *quantum state dispersion* . The simulation name is **quantumdispersion** .

```
Module quantumdispersion
  {
   simulation_name = dispersion1D_el
   regions = all
   quantum_simulation = quantum_el
   min_eigenvalue_number = 0
   max_eigenvalue_number = 5
   wedge = half
   k_space_dimension = 1
   k1 = (0, 0.1, 0)
   number_of_nodes = (10)
   output_format = grace
   plot = k-space_dispersion
  }
```

The dispersion of quantum states is calculated at k-points that are nodes of the mesh in k-space.

The main parameters are:

- **quantum simulation** : name of the efaschroedinger simulation.
- **min eigenvalue number** , **max eigenvalue number** : the dispersion is calculated

for the states number *i* , where

max_eigenvalue_number $\geq i \geq$ min_eigenvalue_number

The rest of the parameters (wedge, k space dimension, etc...) define the k-space.

The output variable name is **k-space_dispersion** . The output format for the dispersion can be controlled independently of the general specification in the `Simulation` section by redefining the `output_format` keyword.

18.1 Module quantumdensity

In Module quantumdensity you can define the calculation of particle (electron,hole) den- sity, based on the result of a previous calculation of the system eigenstates (e.g. with efaschroedinger module). Quantum density may be obtained with an analytical or a numerical calculation.

Analytical calculation of density is done in the following way. For each eigenstate we calculate the effective mass assuming quadratic dispersion. Then the charge density is calculate as follows:

where ρ_{1D} and ρ_{2D} are the 1D and 2D charge densities; m is the averaged mass (the mass is different for each quantized state and is position independent); g is the degeneracy of the states. The + sign is for electrons, the - sign is for holes.

Numerical calculation is done by the following formula:

The integration is performed on a mesh in the k-space.

```
Module quantumdensity
  {
   name = dens_el
   regions = quantum
   plot = quantum_density
   k-space_dimension = 0
   k-space_basis = true
   k1 = (0, 0, 0.1)
   #number_of_nodes = (4)
   #wedge = half
   quantum_simulation = quantum_el
   degeneracy = 2
   refine_fraction = 0.20
   relative_accuracy = 0.01
   refine_k_space = true
   output_density_in_k_space = true
   uniform_refinement = false
   mesh_order = FIRST
   first_state = 3
   initial_eigenstates_number = 10
   analytic = true
  }
```

The available options are:

- **quantum_simulation** : name of the efaschroedinger simulation.
- degeneracy: degeneracy of the quantum state
- **initial_eigenstates_number** : initial number of eigenstates for the Schroedinger

equation

- **analytic** = { true | false } If true then the density is calculated analytically,

otherwise numerically.

18.1.1 Output

The output parameter is **quantum_density**.

18.2 Module opticskp

By defining the Module **opticskp** , calculation of optical properties is enabled; in particular, the optical kp matrix elements are calculated from the quantum models specified in the Module.

The optical spectrum from spontaneous emission is calculated in the following way:

where f_i and f_j are the Fermi distributions.

```
Module opticskp
  {
   name = optics
   regions = quantum
   plot = (optical_spectrum_k_0 )
   initial_state_model = quantum_el
   final_state_model = quantum_hl
   initial_eigenstates = (0, 9)
   final_eigenstates = (0, 15)
   polarization = (0, 0, 1)
   Emin = 2.8
   Emax = 3.6
   dE = 0.001
  }
```

Here, *initial_state_model* and *final_state_model* are, respectively, the quantum simulations (**efaschroedinger** module) associated respectively to the initial state of optical transition (e.g. electron), and to the final state of optical transition (e.g. hole).

initial_eigenstates and *final_eigenstates* refer to the range of eigenstates to be taken in account for optical calculations.

By specifying a range of energy values in this way:

```
Emin = 3.0
Emax = 5.0
dE = 0.001
```

the emission optical spectrum for **k=0** is calculated.

18.2.1 Output

The output variables for optics calculations are:

- **optical_spectrum_k_0** : optical emission spectrum for *k=0*.

18.3 Module opticalspectrum

By defining the Module **opticalspectrum** , optical matrix elements are used to calculate the associated (emission) spectrum with a k-space integration.

```
Module opticalspectrum
  {
   k_space_dimension = 2
   k-space_basis = true
   k1 = (0, 0, 0.1)
   k2 = (0, 0.1, 0)
   refine_fraction = 0.30
   relative_accuracy = 0.01
   refine_k_space = true
   number_of_nodes = (2, 2)
   wedge = quarter
   plot = (optical_spectrum)
   optical_matr_elem_model = opticskp
   polarization = (0, 0, 1)
   Emin = 3.0
   Emax = 5.0
   dE = 0.001
  }
```

The parameters are the following:

`k_space_dimension` = **1** for 2D simulations, **2** for 1D simulations. k-space basis is

true if the k-space is defined by means of k-vectors; if **false** , vectors are expressed in

real space

If `refine_k_space` = **true** , that is adaptive k-mesh refinement is enabled, all the elements whose error is greater than the value (1-refine fraction) (maximum error) are going to be refined. In this case, Error is just the integrated quantity. The refinement will end when the *relative_accuracy* is obtained.

`number_of_nodes` = numb. of elements in k mesh, along each direction

`wedge` = half | quarter, to reduce calculation time, by exploiting symmetry.

`optical_matr_elem_model` = name of the *opticskp* model associated

`polarization` = light polarization (vector)

`Emin, Emax, dE` : energy range and step of spectrum calculation.

18.3.1 Output

The output variables for optics calculations are:

- **optical_spectrum** : k-space integrated optical emission spectrum.

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